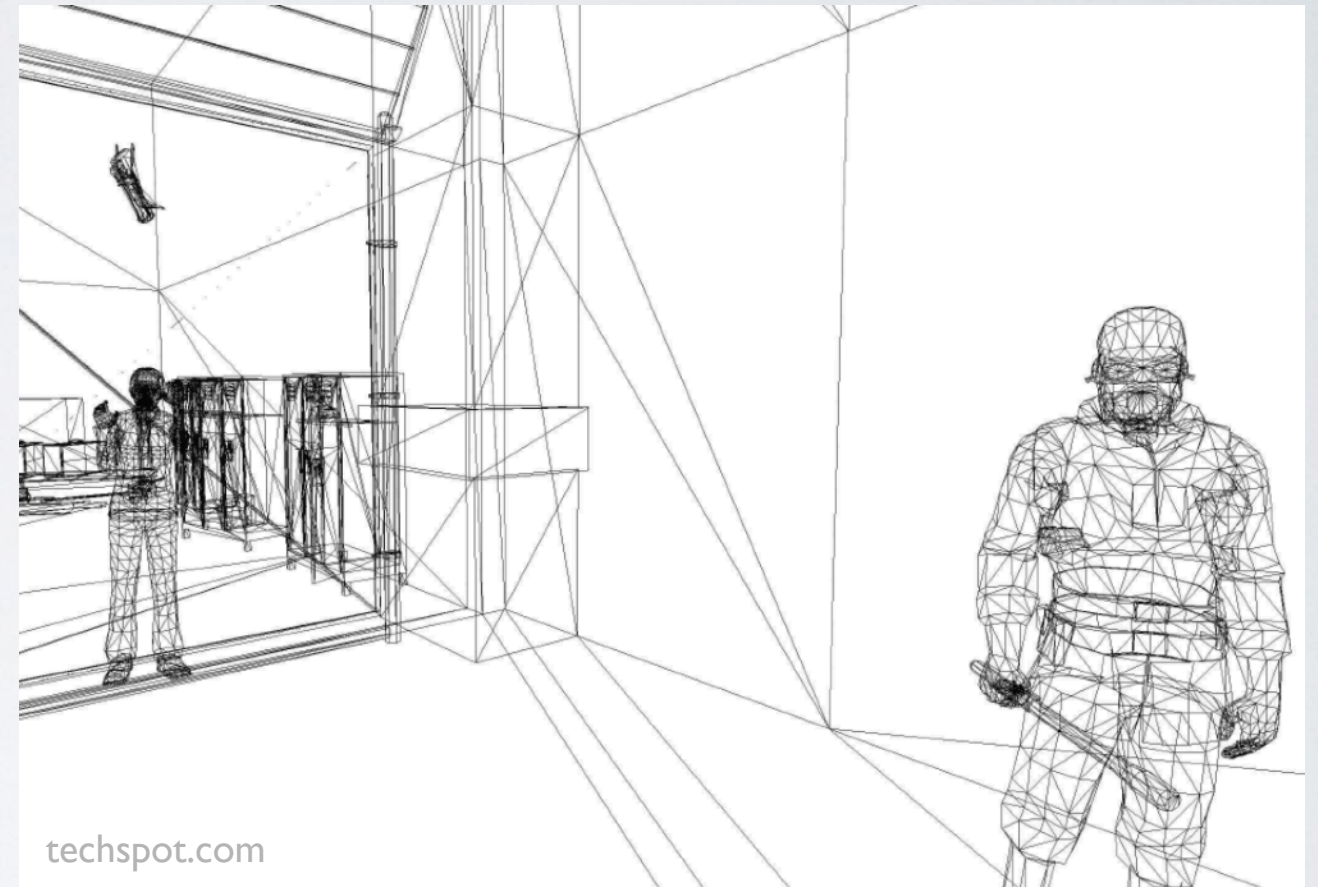
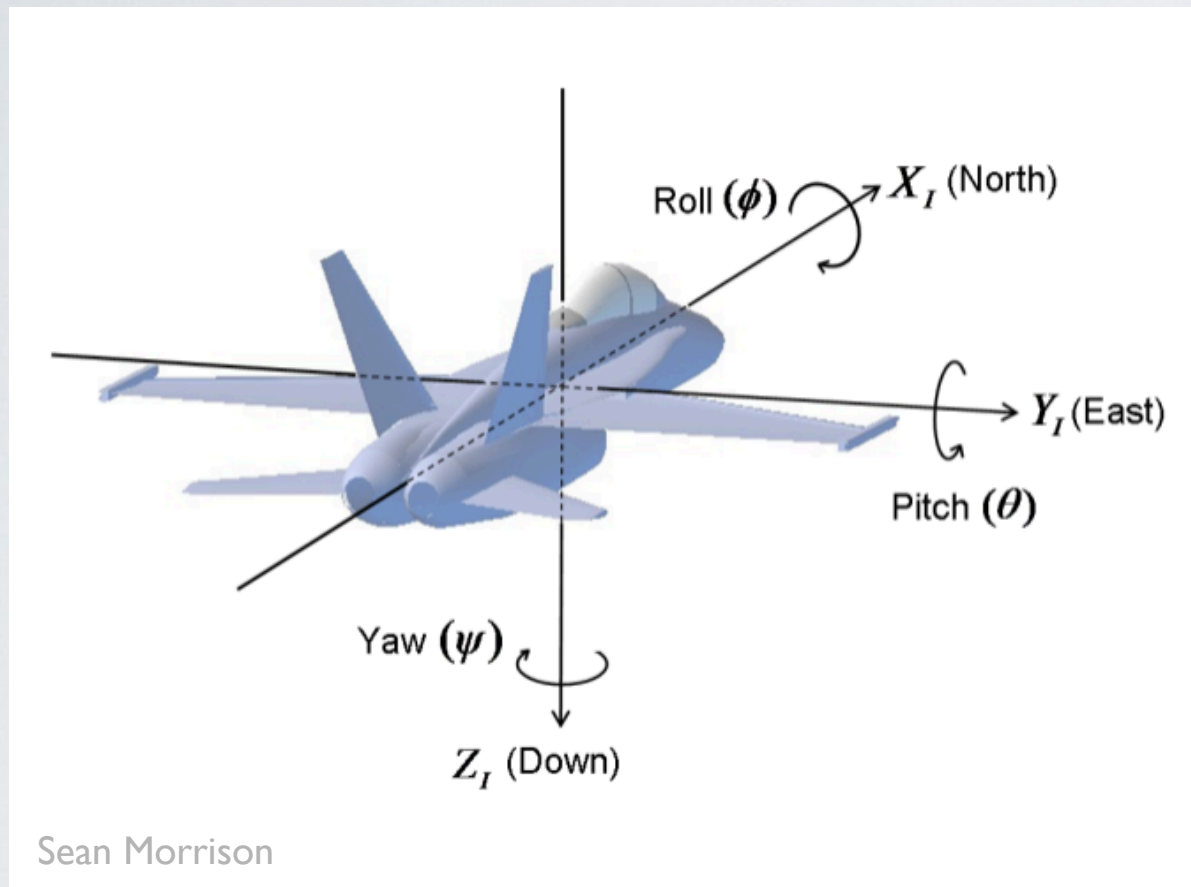


Week 4 - #2

Linear Algebra (IV)



Today: Ch 3

Next Class: Ch 3, 10.1-10.4

Ji-hoon Kim (Seoul National University)

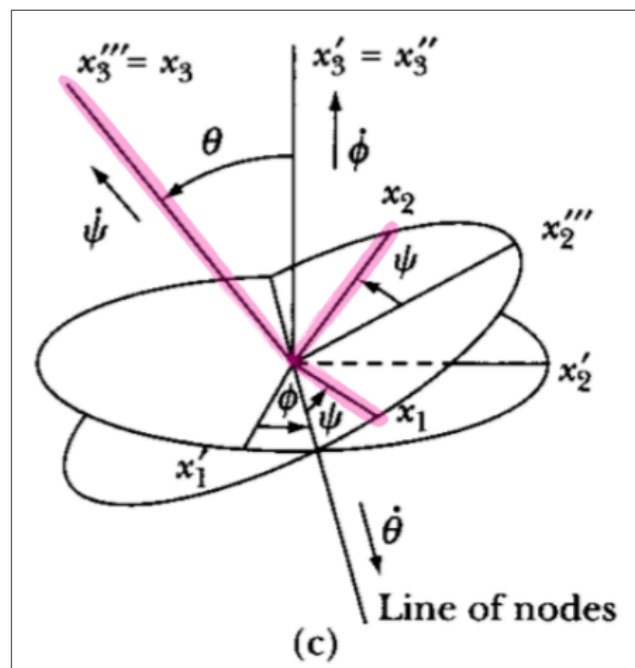
Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #5

— [open book and open note, “and” cellphone or laptop, drop it off as you leave the class] —

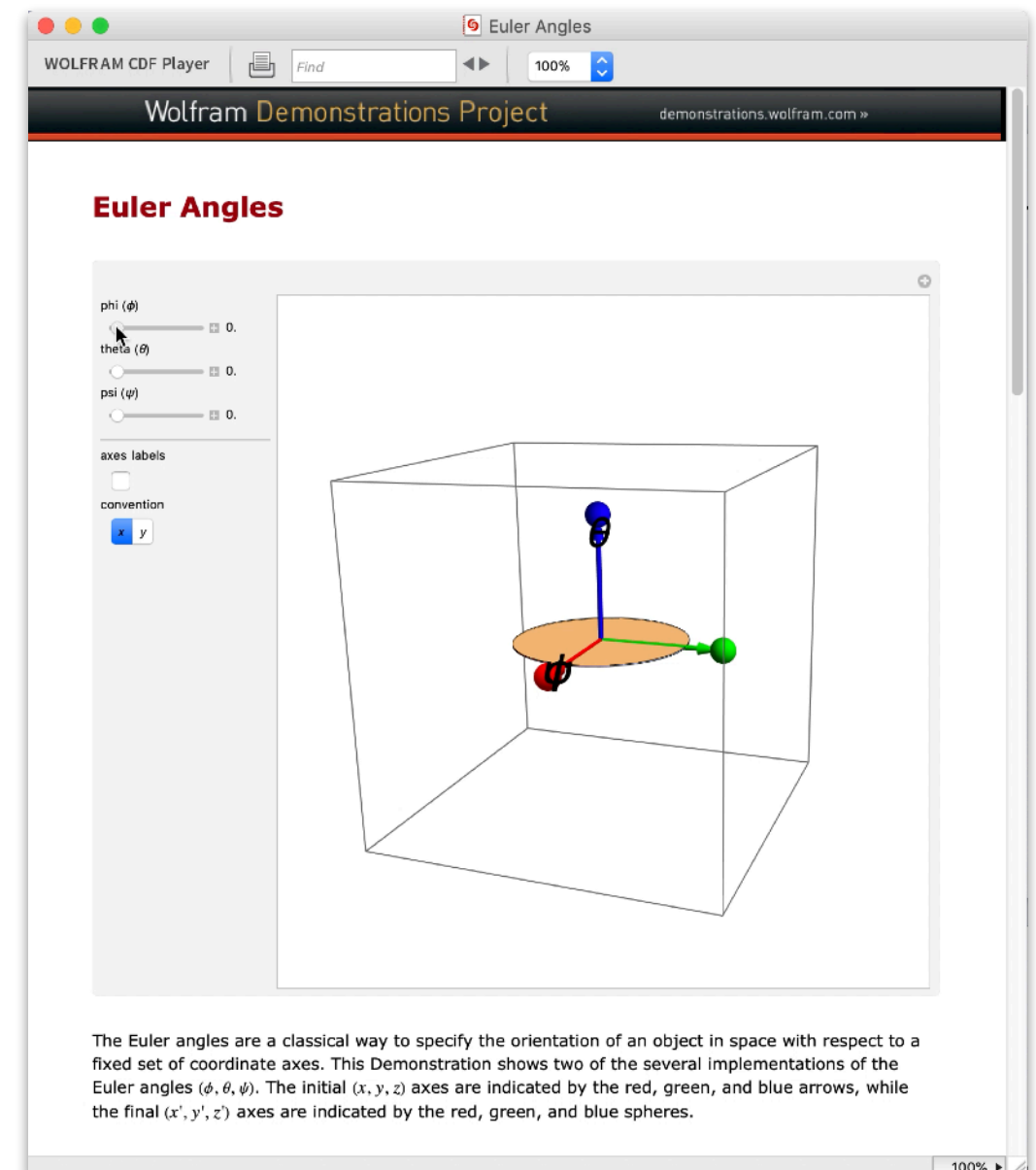
Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Boas, Chapter 3, Section 8, Problem 25
2. Boas, Chapter 3, Section 10, Problem 5(b)
3. Define “conic section” that appears at the beginning of Boas Chapter 3, Section 12.

Thornton & Marion, Figure 11-9(c)



demonstrations.wolfram.com/
EulerAngles/



Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 4, Boas, 3rd ed.

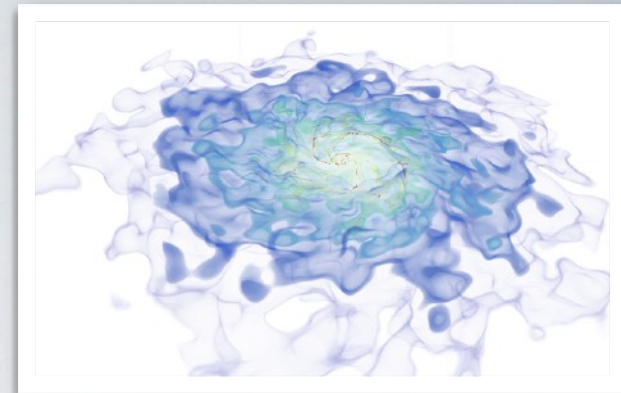
The problems I suggest you to take a deeper look into include, but are not limited to, the following.
The class homework assignments will mainly be from this list.

- Section 01: Problems 7, 8, 9, 15, 16
- Section 02: Problems 1, 2, 3
- Section 04: Problems 2, 3, 7, 15
- Section 05: Problems 1, 2, 6, 8
- Section 06: Problems 1, 4, 9
- Section 07: Problems 2, 4, 7, 10, 16, 25, 28
- Section 08: Problems 2, 8, 11, 13
- Section 09: Problems 2, 5, 9, 11
- Section 10: Problems 2, 5, 10, 14
- Section 11: Problems 1, 4, 6, 7, 8
- Section 12: Problems 2, 5, 10, 13, 16, 19
- Section 13: Problems 6, 9, 16, 27, 29

Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 5, Boas, 3rd ed.

The problems I suggest you to take a deeper look into include, but are not limited to, the following.
The class homework assignments will mainly be from this list.

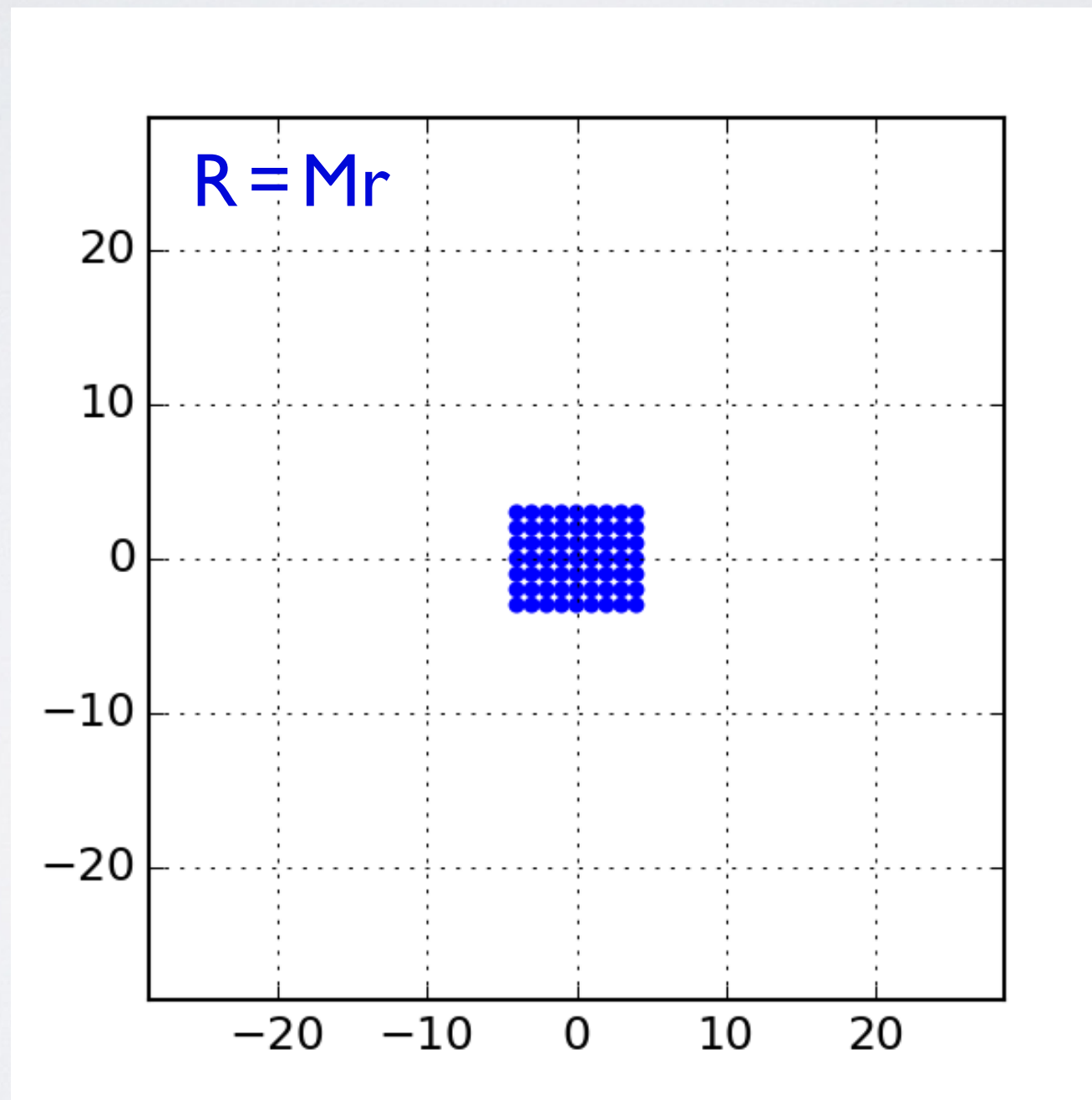
- Section 01: Problems 2, 4, 5
- Section 02: Problems 3, 5, 9, 11, 13, 16, 20, 22, 26, 39, 41
- Section 03: Problems 7, 12, 13, 17, 18, 19, 20, 21, 25
- Section 04: Problems 2, 4, 7, 14, 16, 19, 20
- Section 05: Problems 1, 5
- Section 06: Problems 4, 6, 11, 18, 23, 25, 27



Meaning of Diagonalization

Meaning of Diagonalization (Ch 3.1 I)

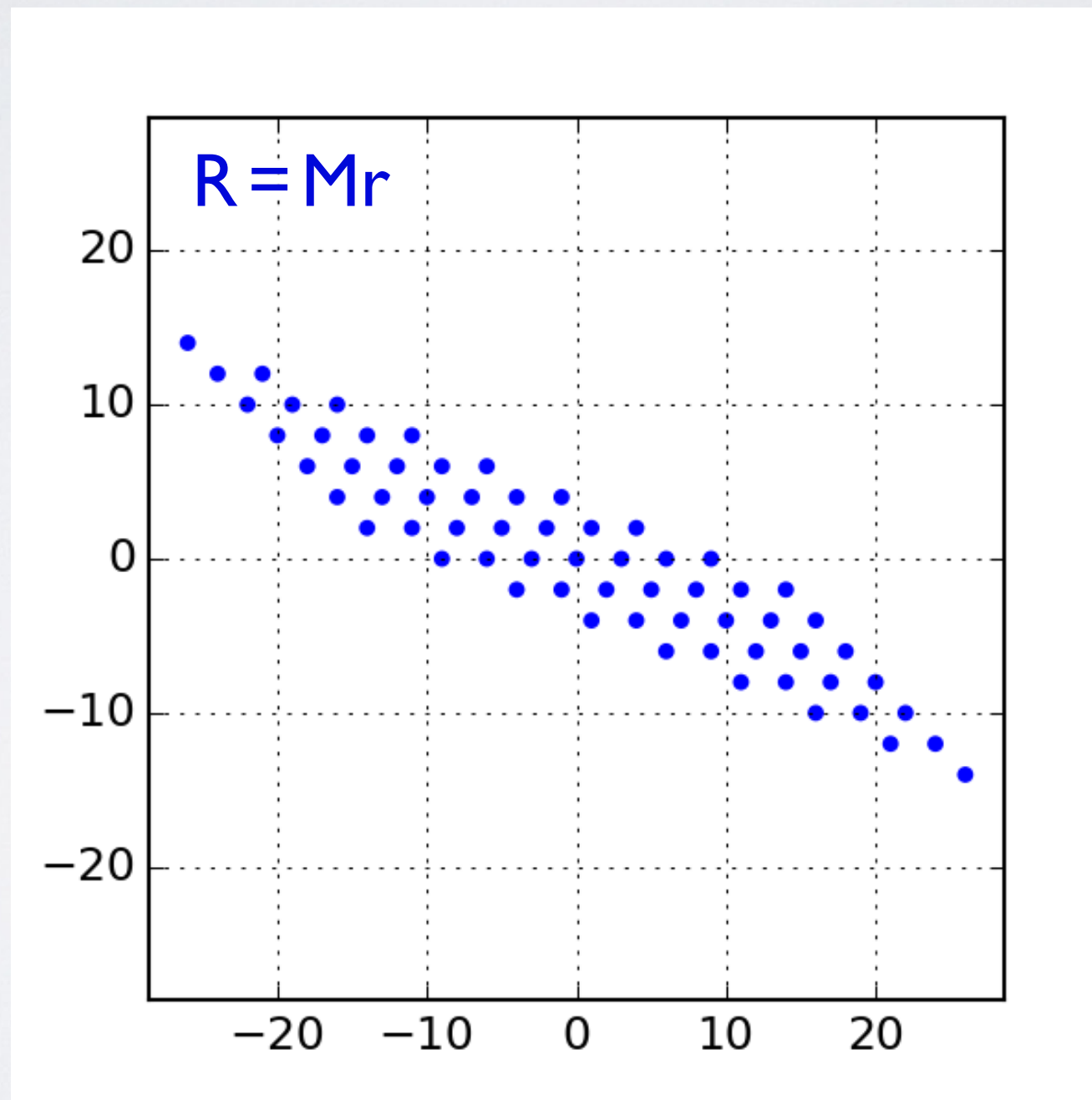
- Matrix C **changes** the old set of coordinate axes (x, y) to a new, convenient set of coordinate axes (x', y') .



modified from dododas.github.io/linear-algebra-with-python/posts/16-12-29-2d-transformations.html

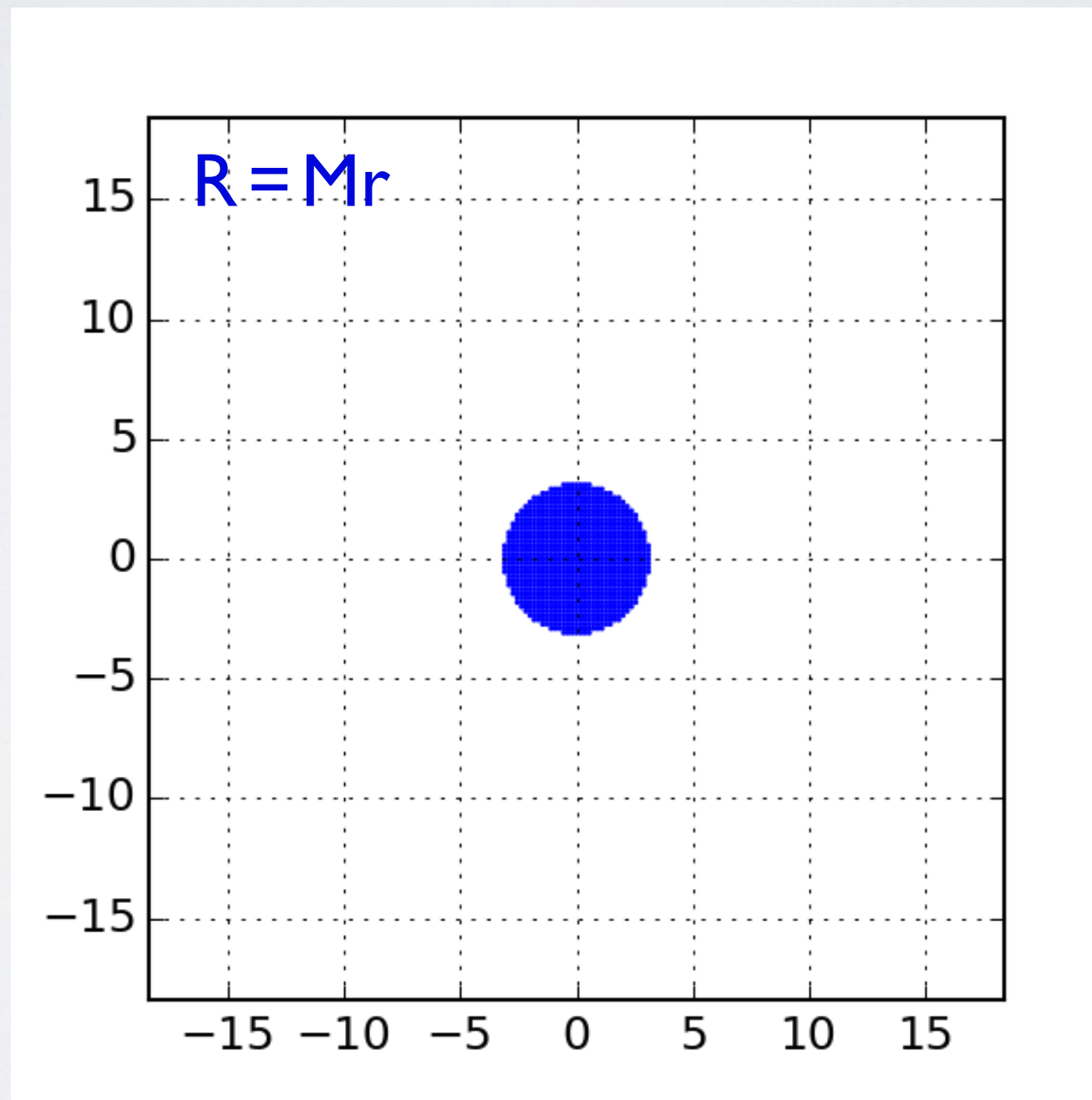
Meaning of Diagonalization (Ch 3.1 I)

- Matrix C **changes** the old set of coordinate axes (x, y) to a new, convenient set of coordinate axes (x', y') .



Meaning of Diagonalization (Ch 3.1 I)

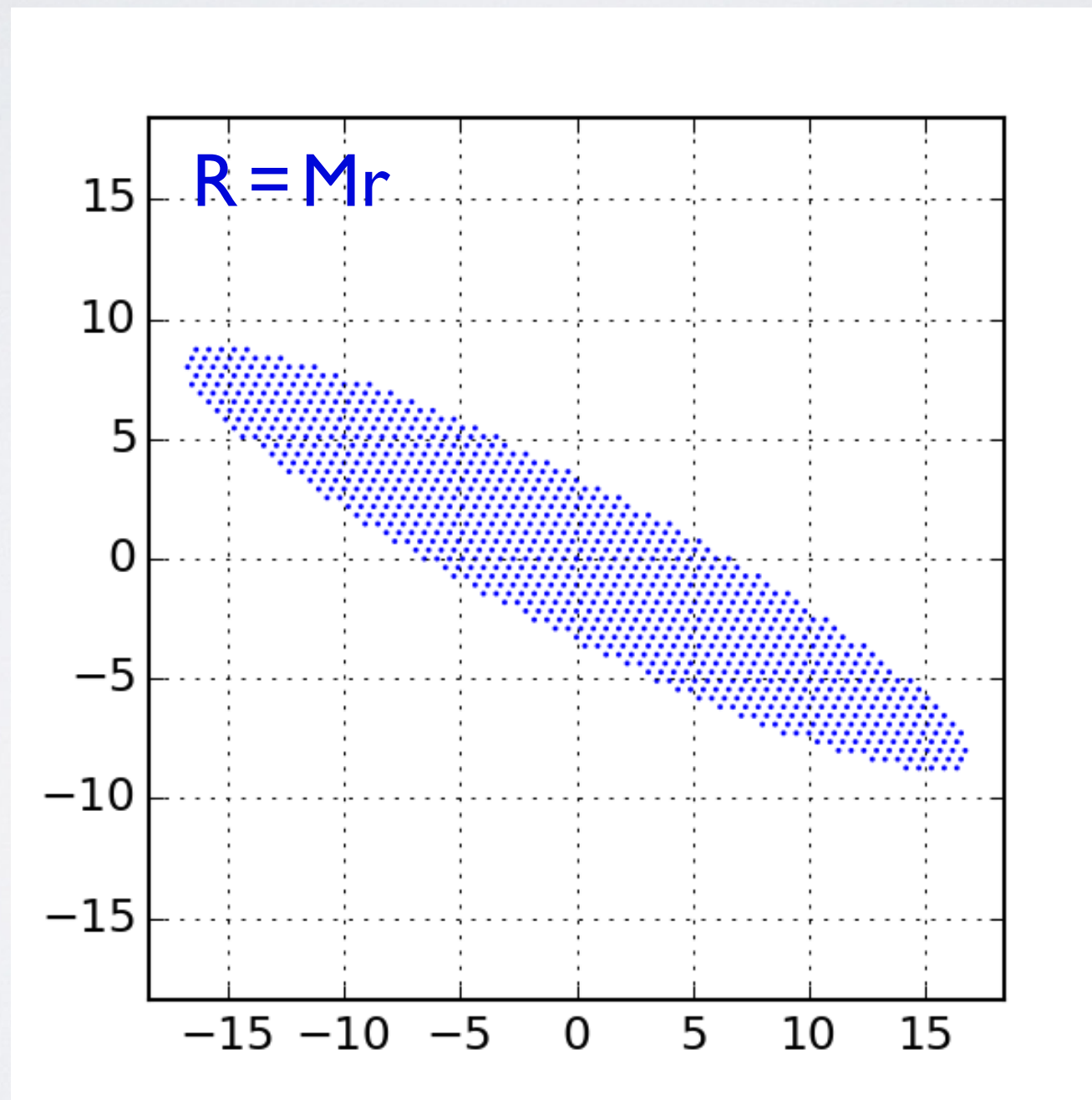
- Matrix C **changes** the old set of coordinate axes (x, y) to a new, convenient set of coordinate axes (x', y') .



modified from dododas.github.io/linear-algebra-with-python/posts/16-12-29-2d-transformations.html

Meaning of Diagonalization (Ch 3.1 I)

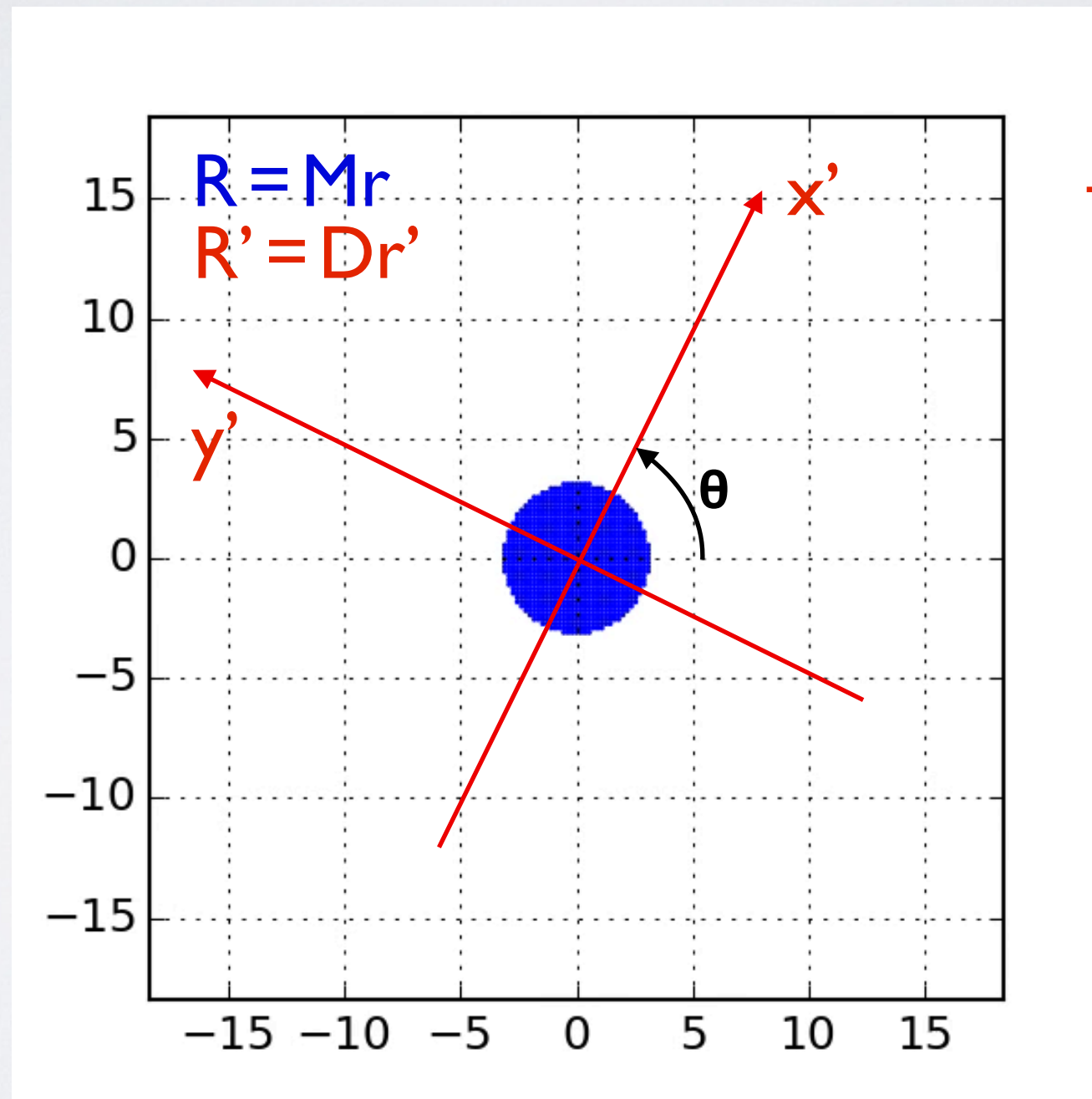
- Matrix C **changes** the old set of coordinate axes (x, y) to a new, convenient set of coordinate axes (x', y') .



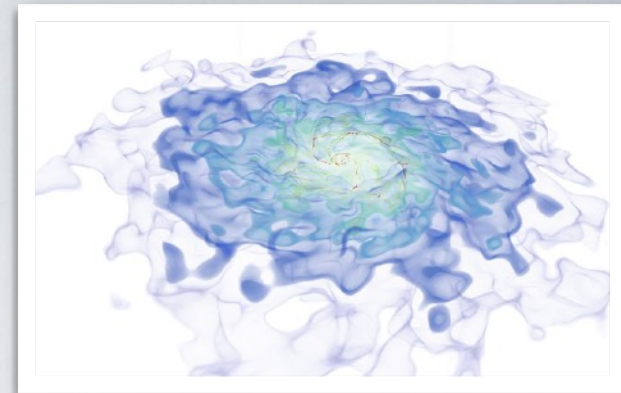
modified from dododas.github.io/linear-algebra-with-python/posts/16-12-29-2d-transformations.html

Meaning of Diagonalization (Ch 3.1 I)

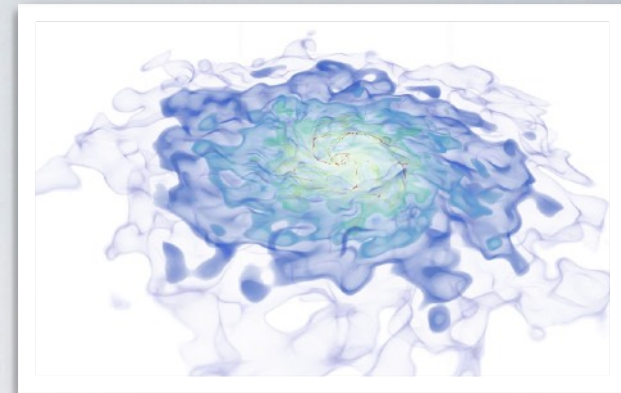
- Matrix C **changes** the old set of coordinate axes (x, y) to a new, convenient set of coordinate axes (x', y') .



→ Figure 11.3



A Few Logistics Items

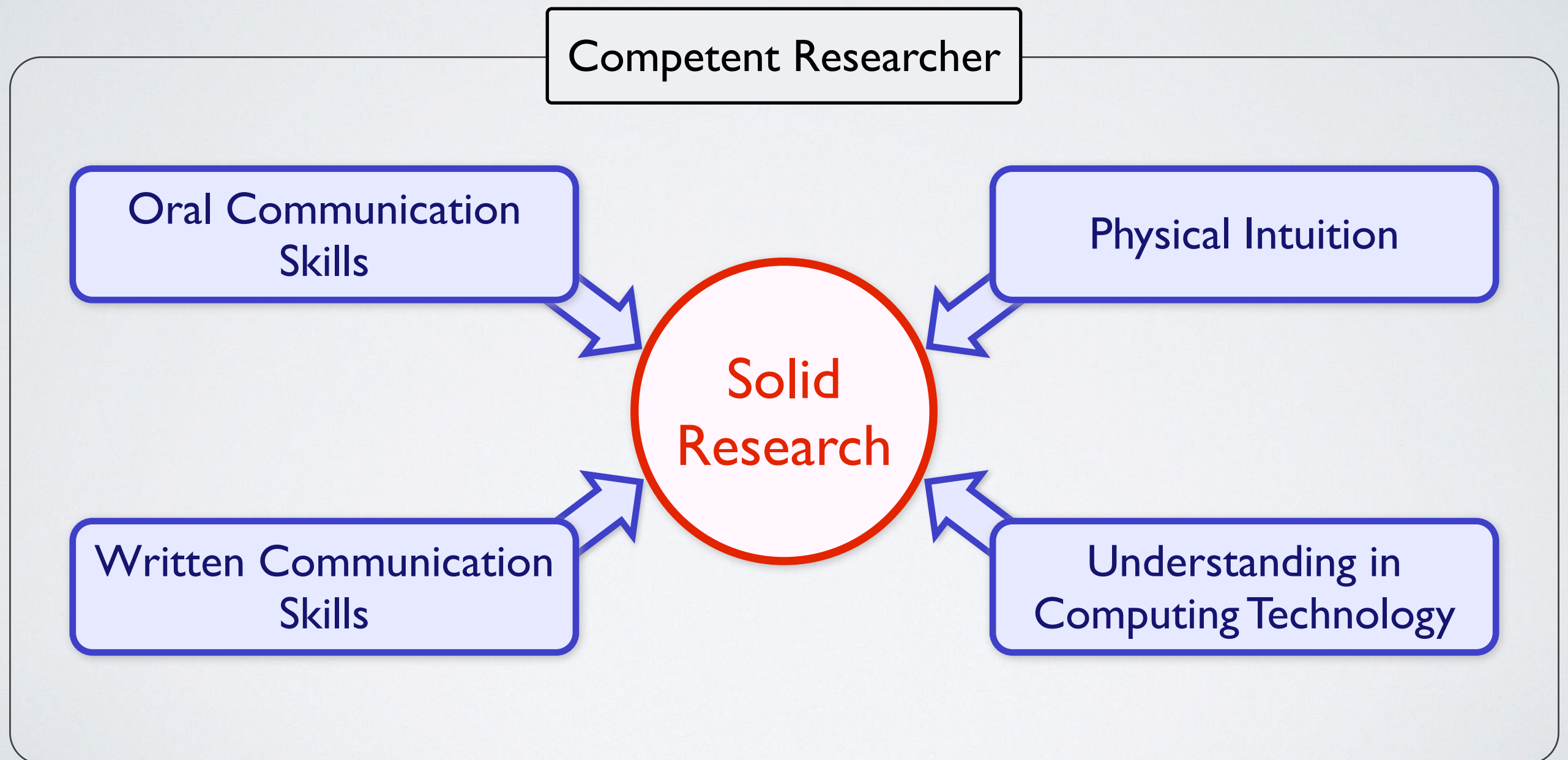


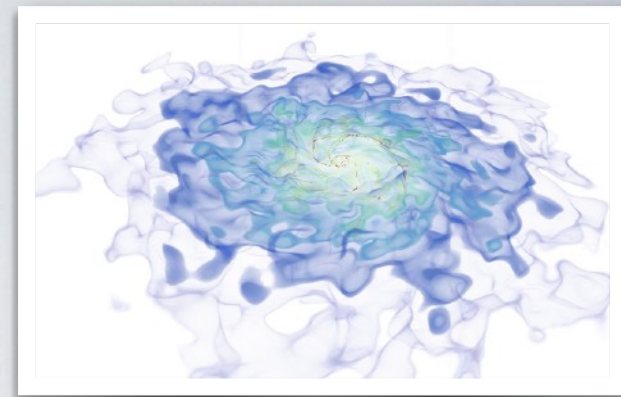
Essential Skillsets For Future Physicists & Astrophysicists

(on top of the strong physics/math background you will build)

Essential Skillsets For (Astro)Physicists

- To become proficient and effective researchers while **relying less on guesswork as you navigate through the academia**





Term Project Paper

Grading Scheme

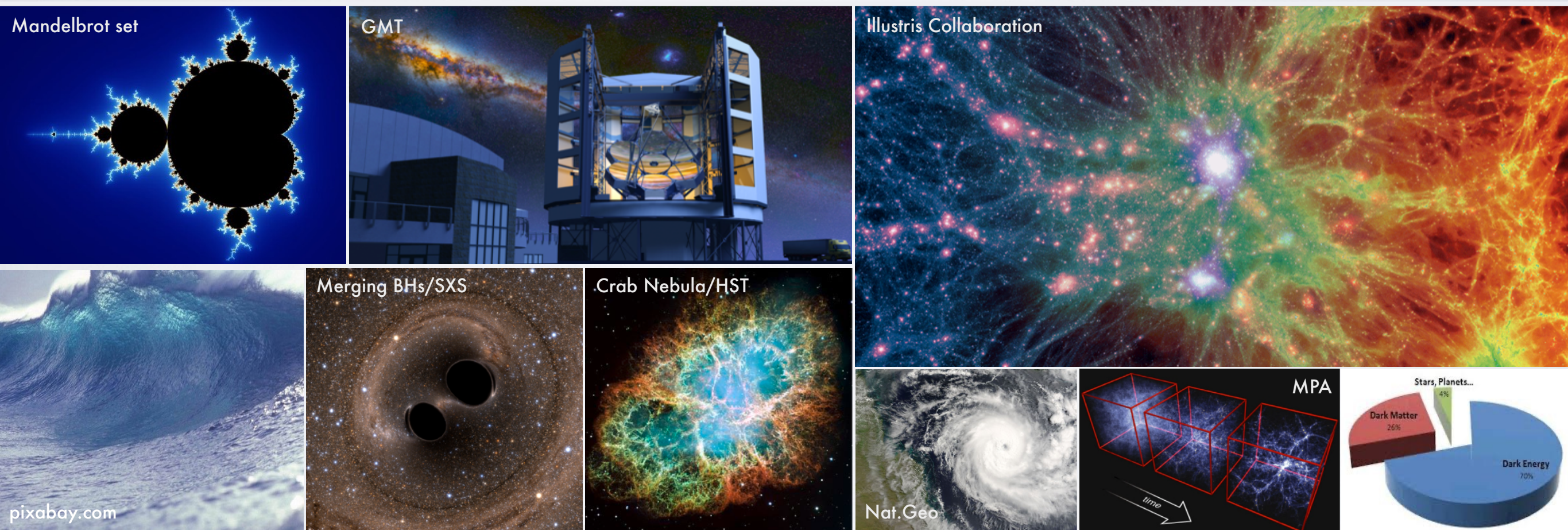
- Grade policy in the course syllabus.

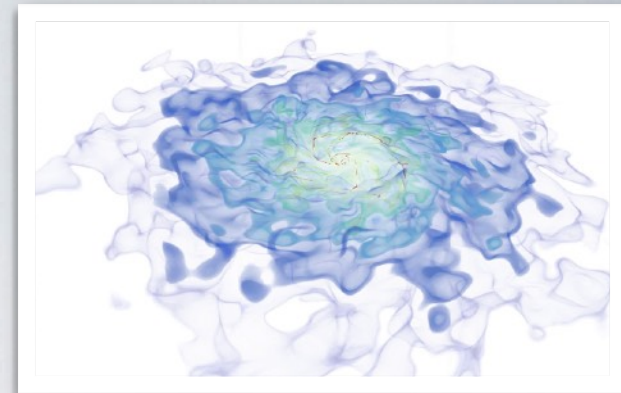
* 3. 평가방법	구분	출석	과제	중간	기말	수시평가	태도	기타	합계
	비율	5	35	20	20	15	5	0	100
	비고			지필고사	지필고사				
	출석 규정 : 수업일수의 1/3을 초과하여 결석하면 성적은 "F" 또는 "U"가 됨(학칙 85조) (담당교수가 불가피한 결석으로 인정하는 경우는 예외로 할 수 있음)								

- 6 HW assignments - Scheduled in 3, 5, 7, 10, 12, 14th week
 - Due in <1 weeks on eTL or in the classroom
 - Only handwritten answers accepted (except numerical problems)
- Term project paper - Due near the end of the semester
 - To be discussed in detail later

Term Project Paper

- Explore how ideas and methods from mathematics and mathematical physics play a role in modern research or everyday phenomena. Must approach the topic creatively and **incorporate your own original insights**.
- Due: Before the class starts on **Nov. 30**
(**10 page-limit** including cover/reference; typewritten report okay)



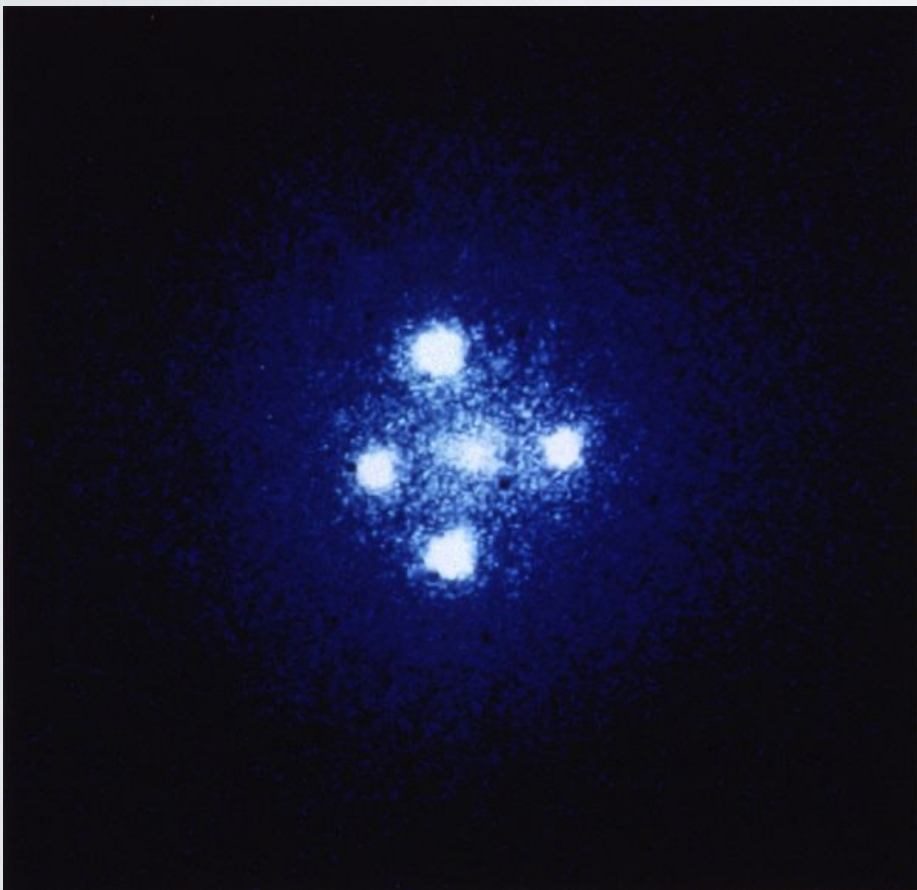


My Examples This Semester

Strong Lensing Regime

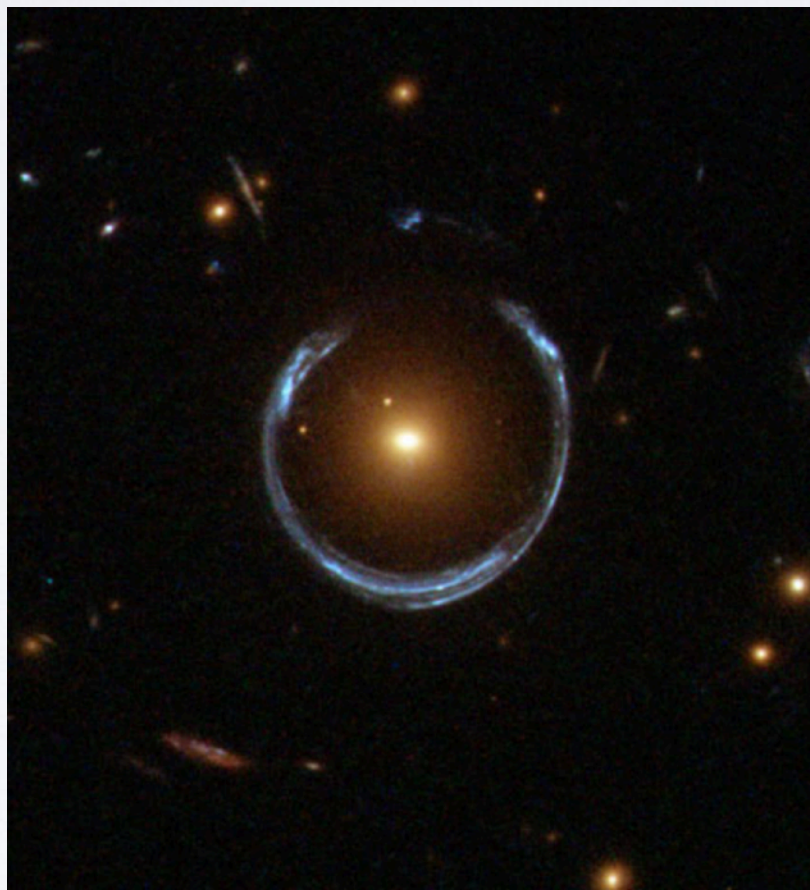
- Effect **strong enough** to be noticeable by eyes: multiple images like Einstein crosses, rings and arcs (distortion $>$ intrinsic size).

quasar - galaxy lensing



QSO2237+0305, NASA/HST

galaxy - galaxy lensing



LRG 3-757, NASA/HST

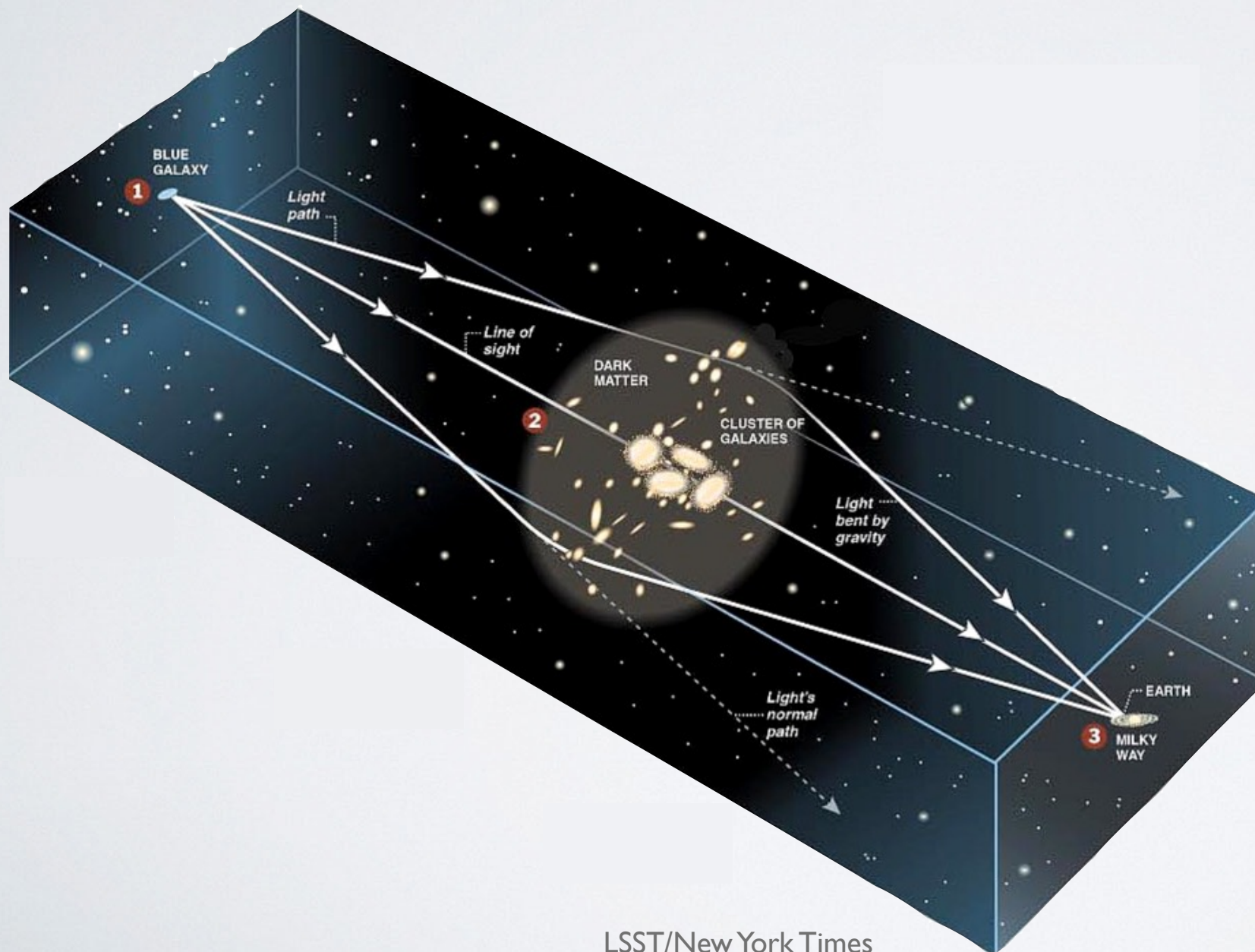
galaxy - galaxy cluster lensing



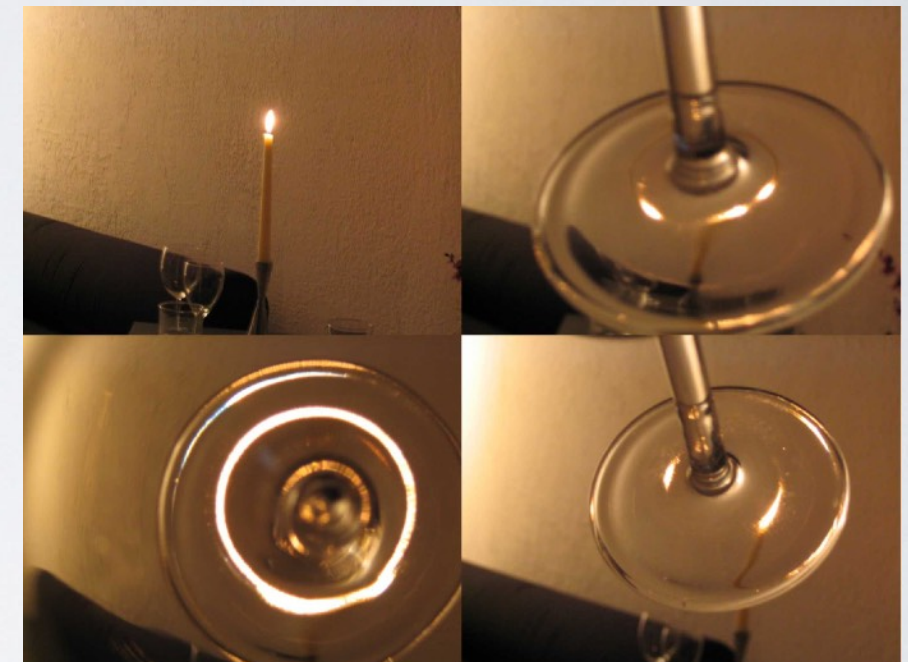
"Smiley", SDSS J1038+4849, NASA/HST

Gravitational Lens As A Telescope

- Galaxy clusters act as **nature-made cosmic telescopes** allowing us to observe the young Universe (~ 10 to 10^3 times magnified).



LSST/New York Times



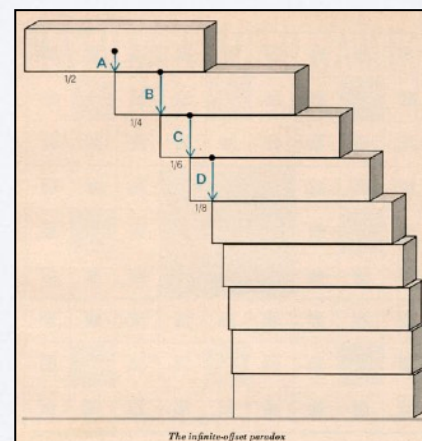
Marshall

Fun with Stacking Blocks

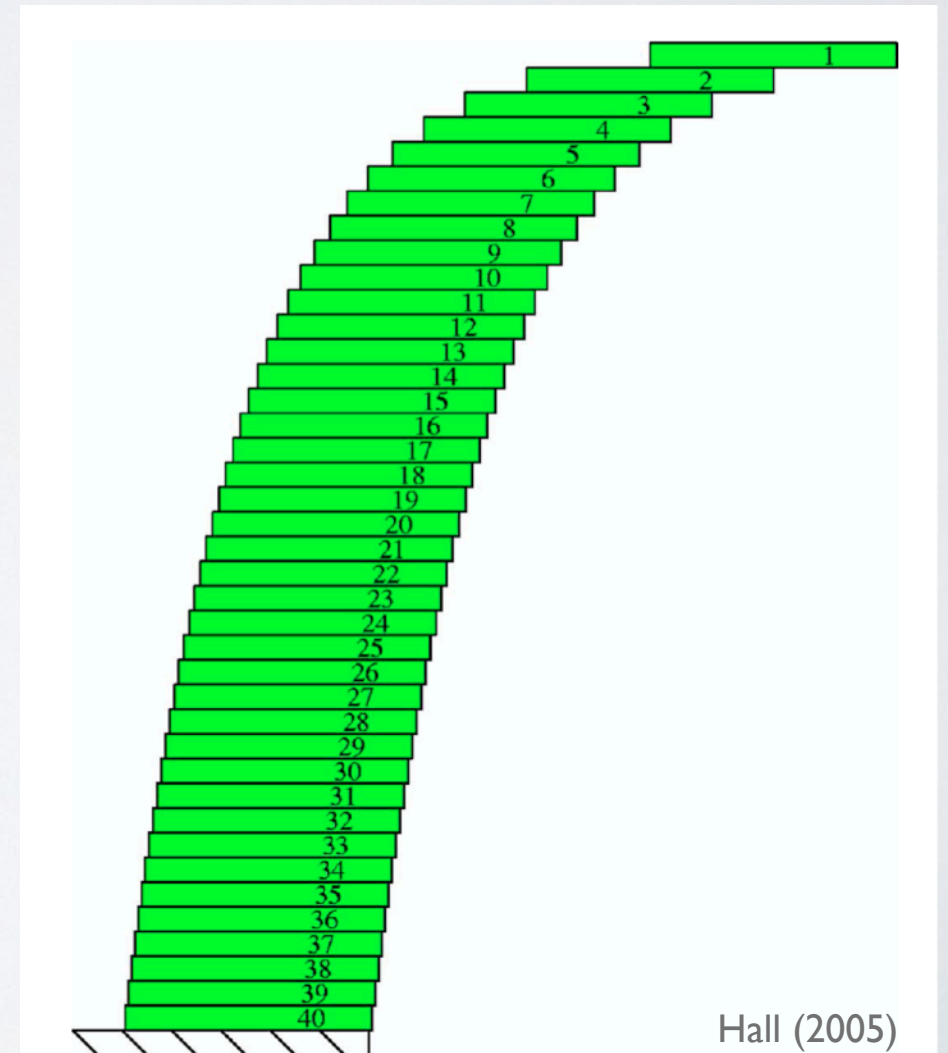
- See: Problem 16.1 + “Fun with Stacking Blocks” (Hall, 2005), etc.



N=13



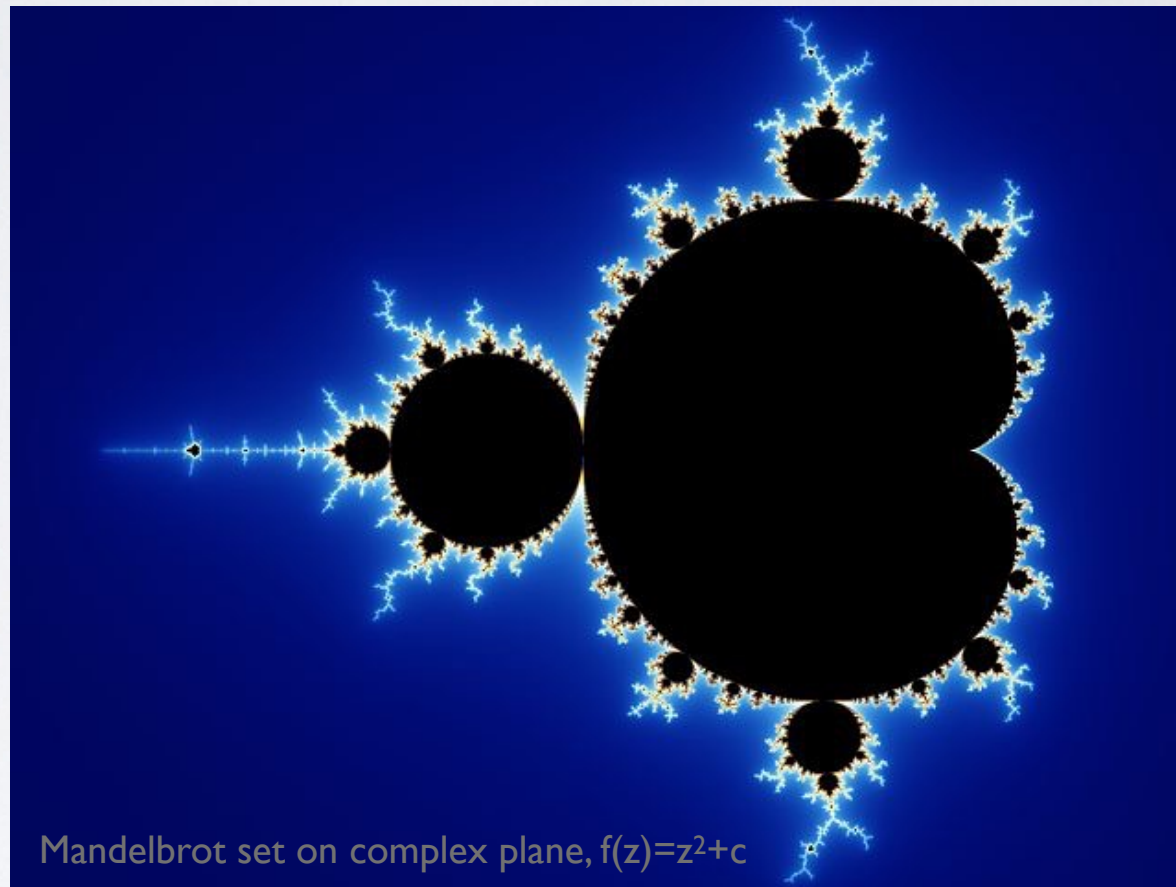
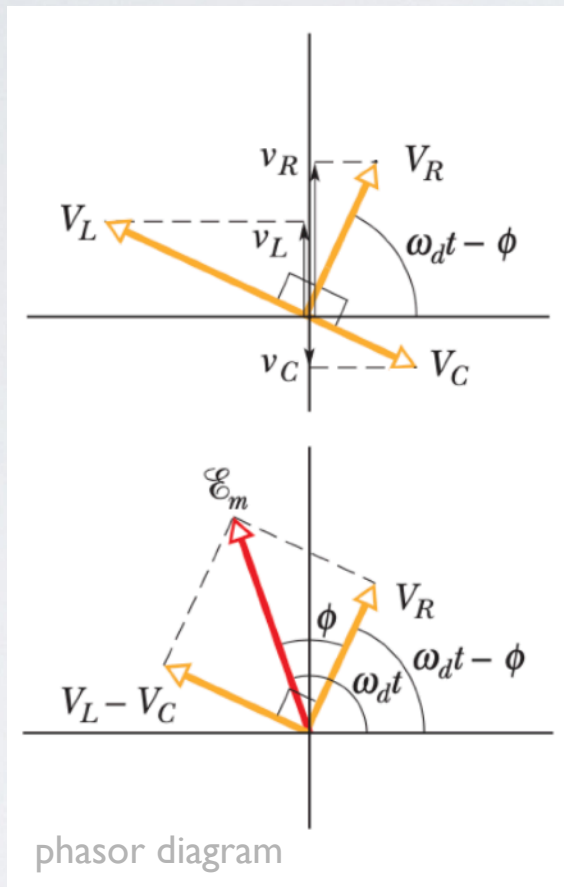
bit-player.org



Hall (2005)

Week 2 - #2

Complex Numbers (II)



Today: Ch 1-2

Next Class: Ch 2

Ji-hoon Kim (Seoul National University)

Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #2

— [open book and open note, “and” cellphone or laptop, drop it off as you leave the class] —

Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Define “complex plane” from your textbook; then, Boas, Chapter 2, Section 5, Problem 59
2. Briefly explain the so-called Mandelbrot set demonstrated in the following movie — in particular, how this one of the most famous images in the fractal/chaos theory is plotted on a “complex plane”.



youtube.com/watch?v=G_GBwuYuOOs,
Mandelbrot set on complex plane

Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #6

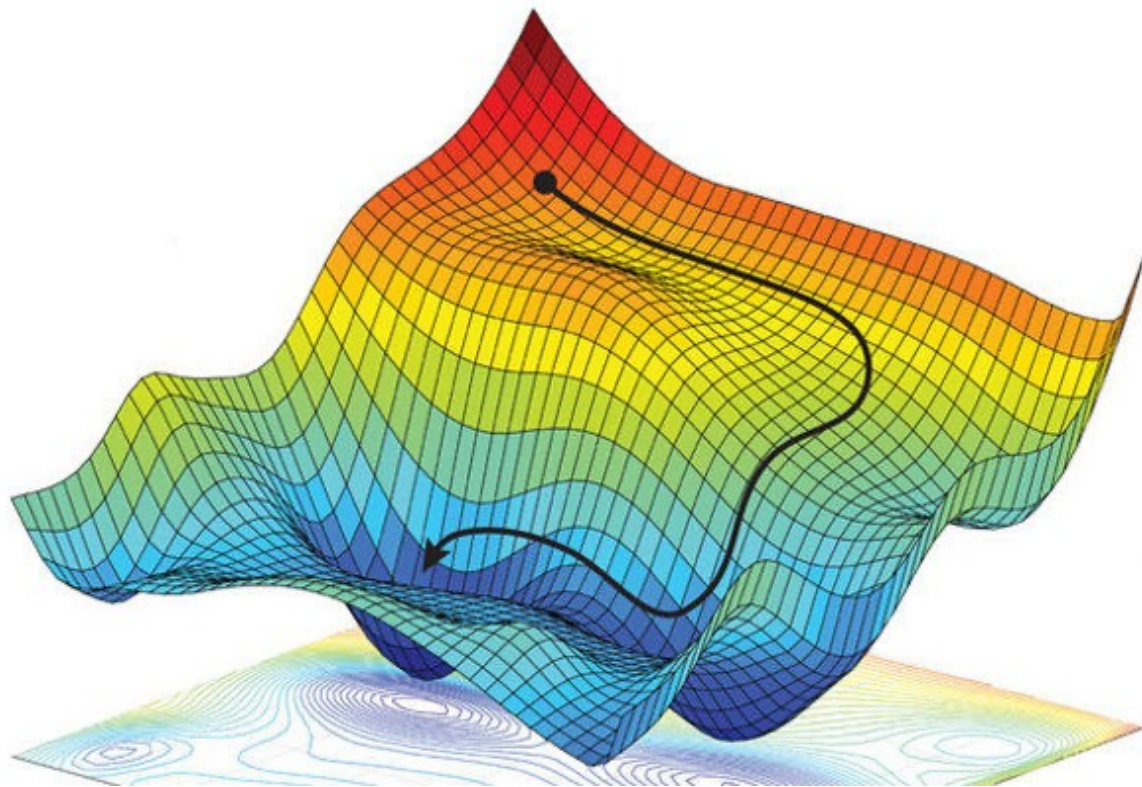
— [open book and open note, “and” cellphone or laptop, drop it off as you leave the class] —

Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

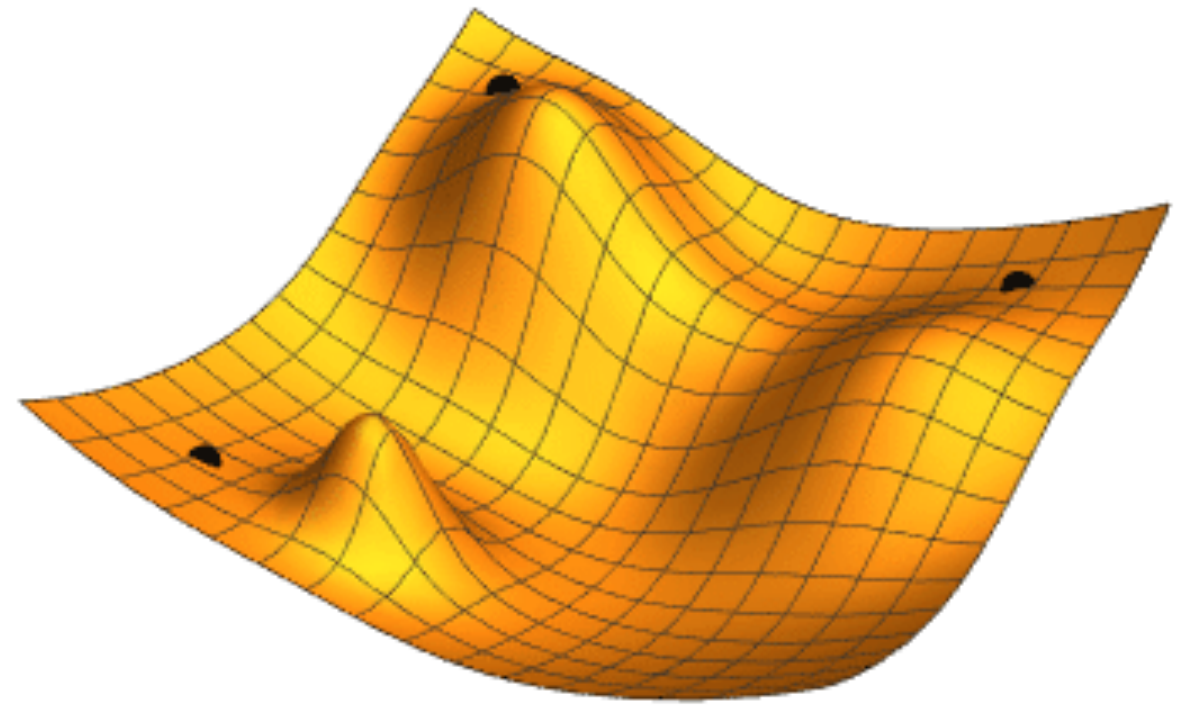
1. Boas, Chapter 3, Section 11, Problem XX

2. Study and briefly explain how matrix operations are used in 3-dimensional rendering in various fields such as game design, immersive visualization, augmented reality, and others.

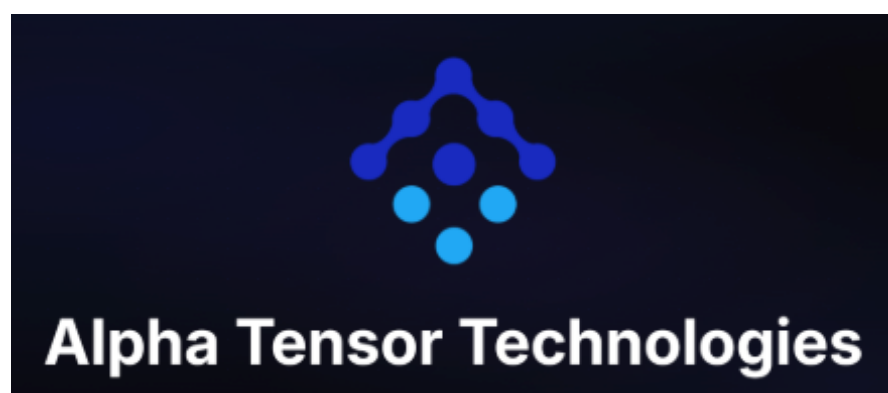




Amini et al. (2018)

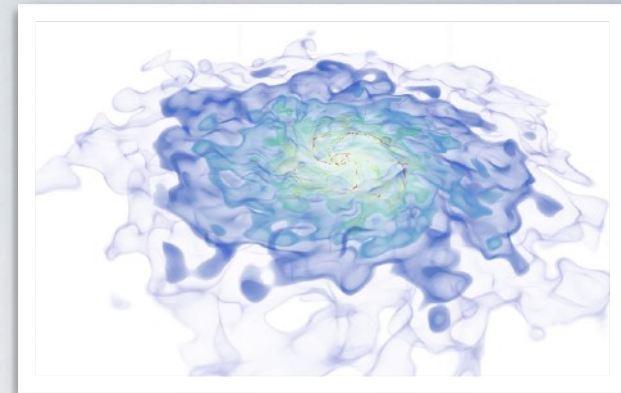


gradient descent, Wikipedia commons



[tensorflow.org](https://www.tensorflow.org)

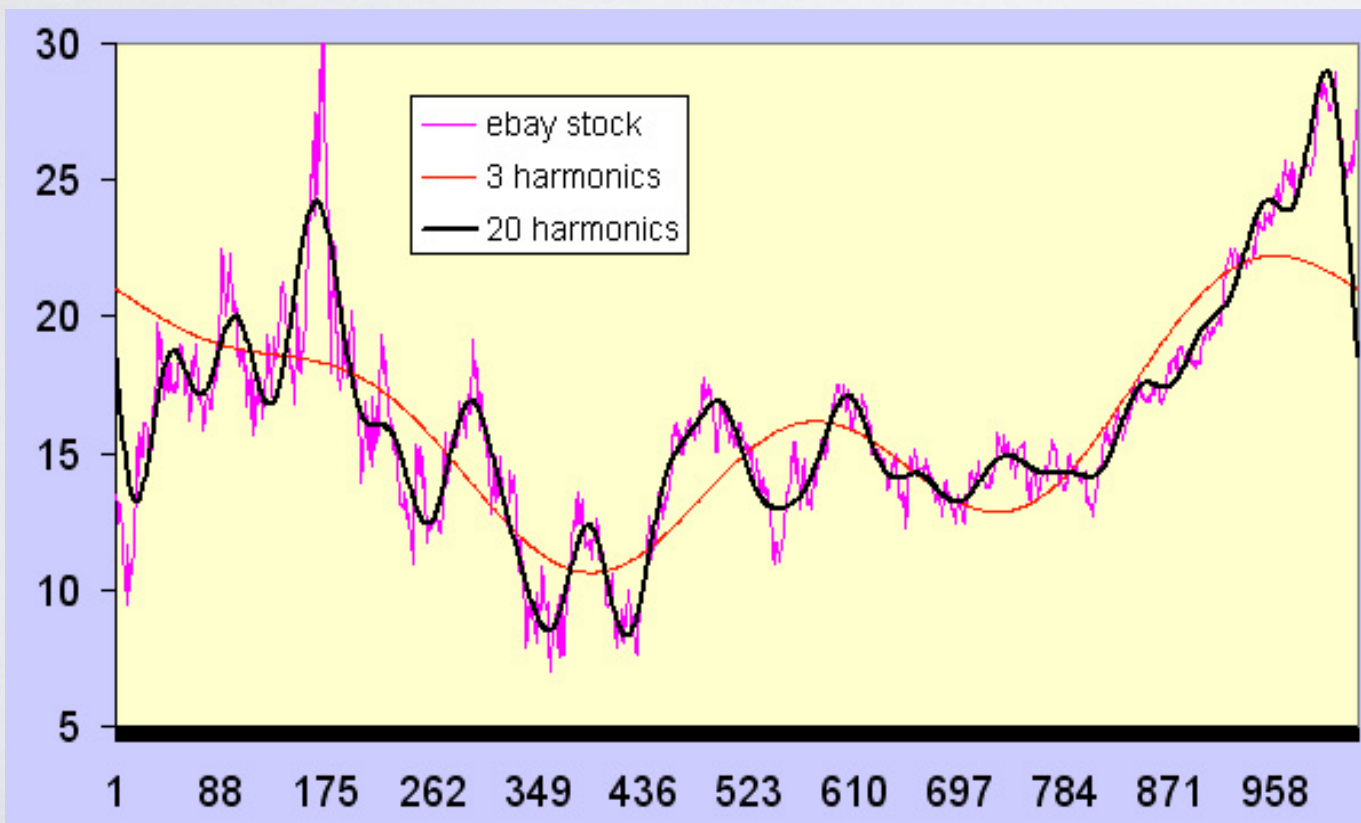




Project Ideas

Term Project Paper

- Explore how ideas and methods from mathematics and mathematical physics play a role in modern research or everyday phenomena. Must approach the topic creatively and **incorporate your own original insights**.
- Due: Before the class starts on **Nov. 30**
(**10 page-limit** including cover/reference; typewritten report okay)



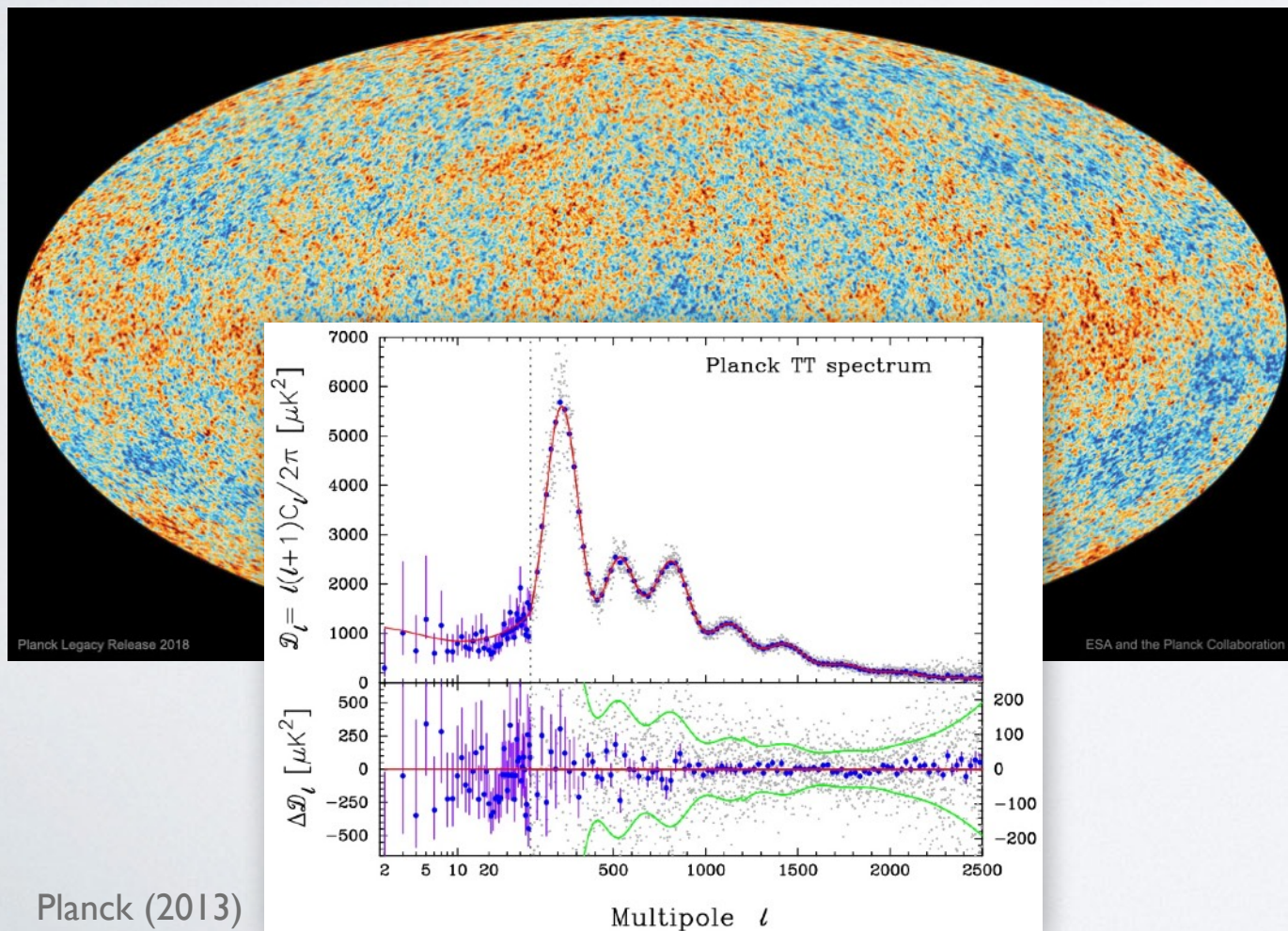
intelligenttradingtech.blogspot.com

Grading guideline

- 0: no paper found with your name on it
- 1: paper found with your name on it
- 2: some scribbles
- 3: reasonable problem & solution
- 4: unique, interesting problem & solution
- 5: reserved for those exceptional

Term Project Paper

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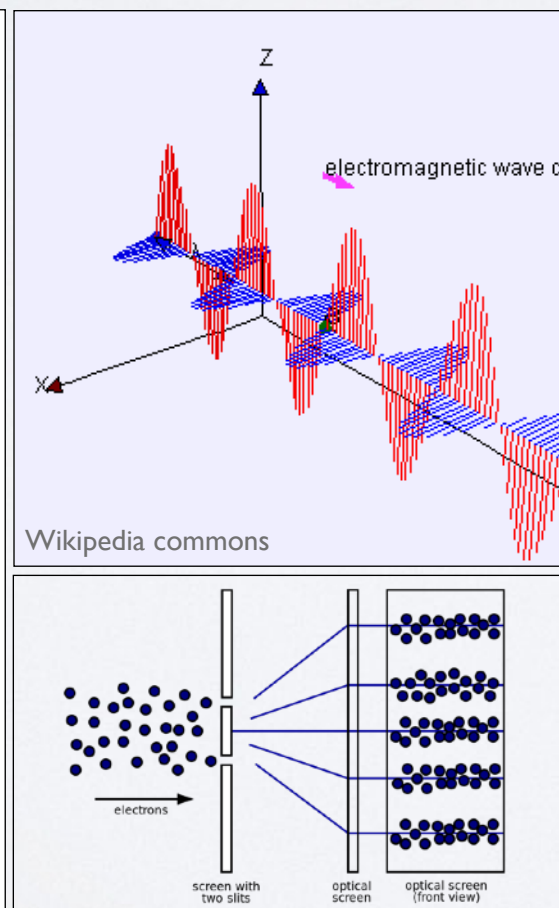
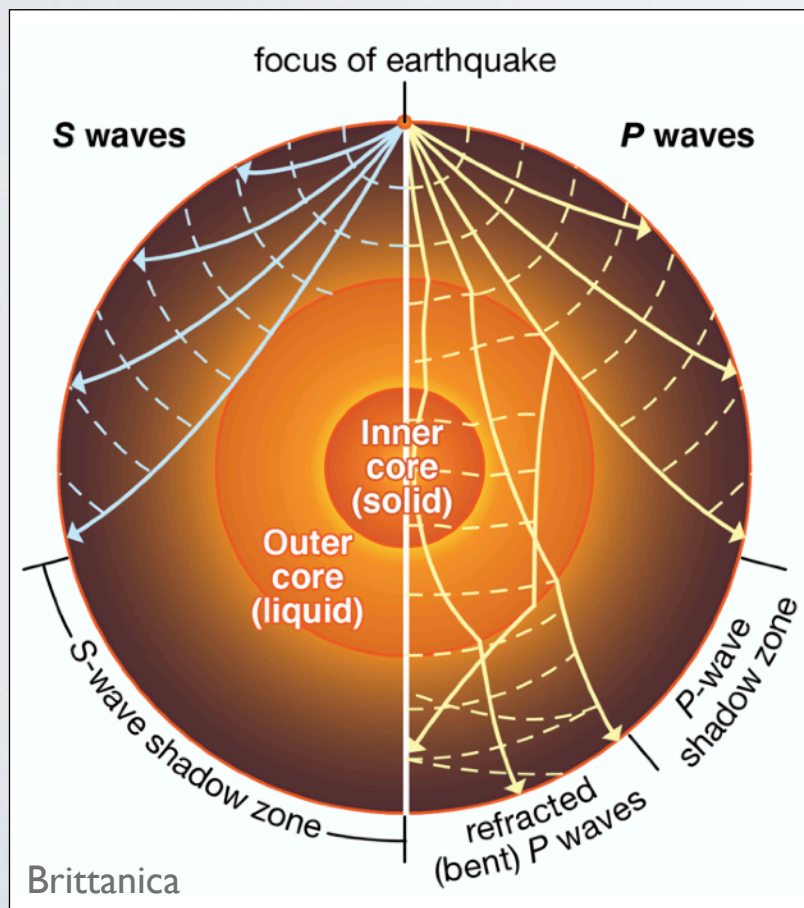
Planck (2013)

Grading guideline

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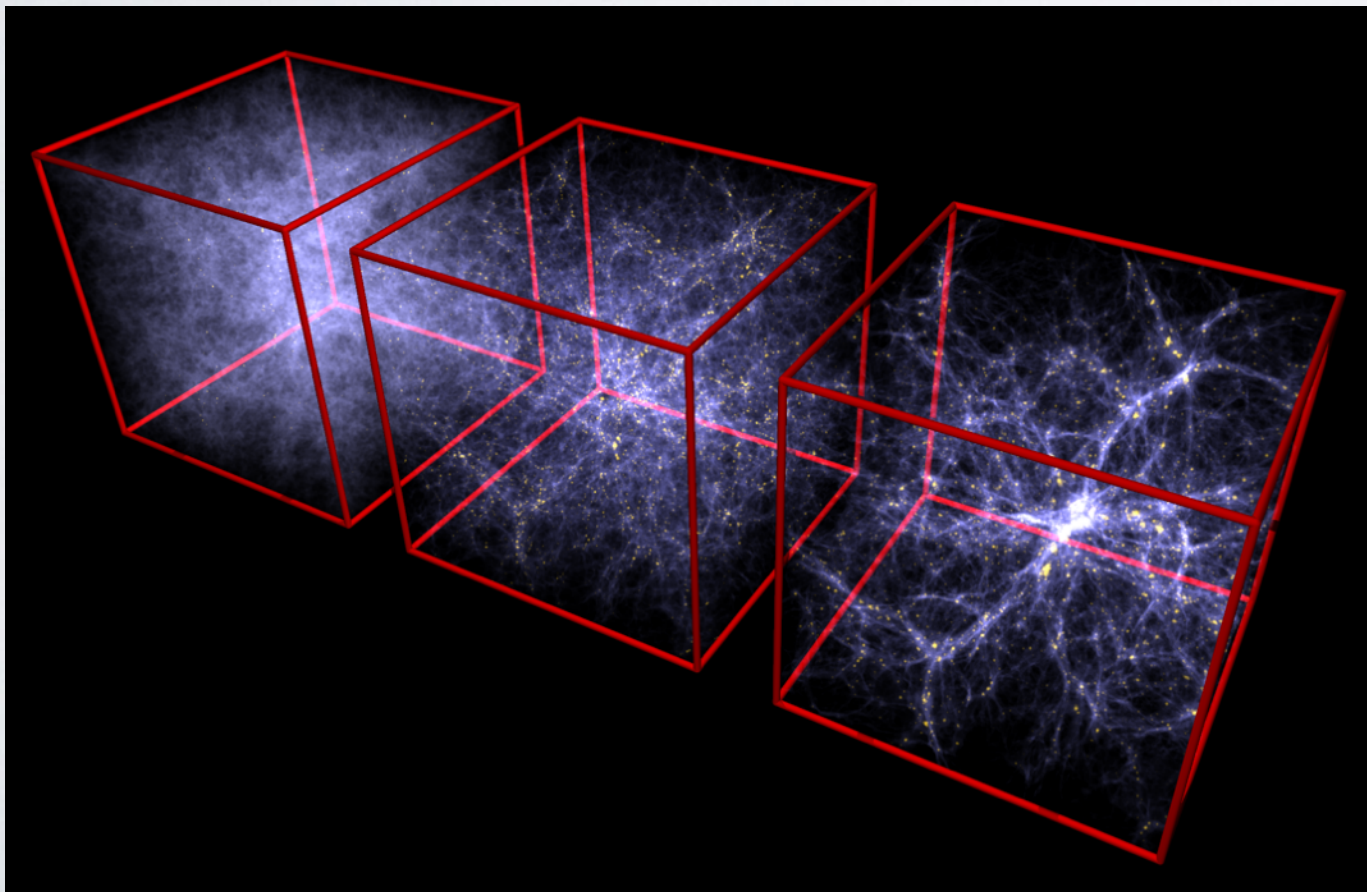


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(**10 page-limit** including cover/reference; typewritten report okay)



Structure Formation, MPA

Grading guideline

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Research

Discovering novel algorithms with AlphaTensor

October 5, 2022

Article

Discovering faster matrix multiplication algorithms with reinforcement learning

<https://doi.org/10.1038/s41586-022-05172-4>

Received: 2 October 2021

Accepted: 2 August 2022

Published online: 5 October 2022

Open access

Check for updates

Alhussein Fawzi^{1,2,3}, Matej Balog^{1,2}, Aja Huang^{1,2}, Thomas Hubert^{1,2}, Bernardino Romera-Paredes^{1,2}, Mohammadamin Barekatain¹, Alexander Novikov¹, Francisco J. R. Ruiz¹, Julian Schrittwieser¹, Grzegorz Swirszcz¹, David Silver¹, Demis Hassabis¹ & Pushmeet Kohli¹

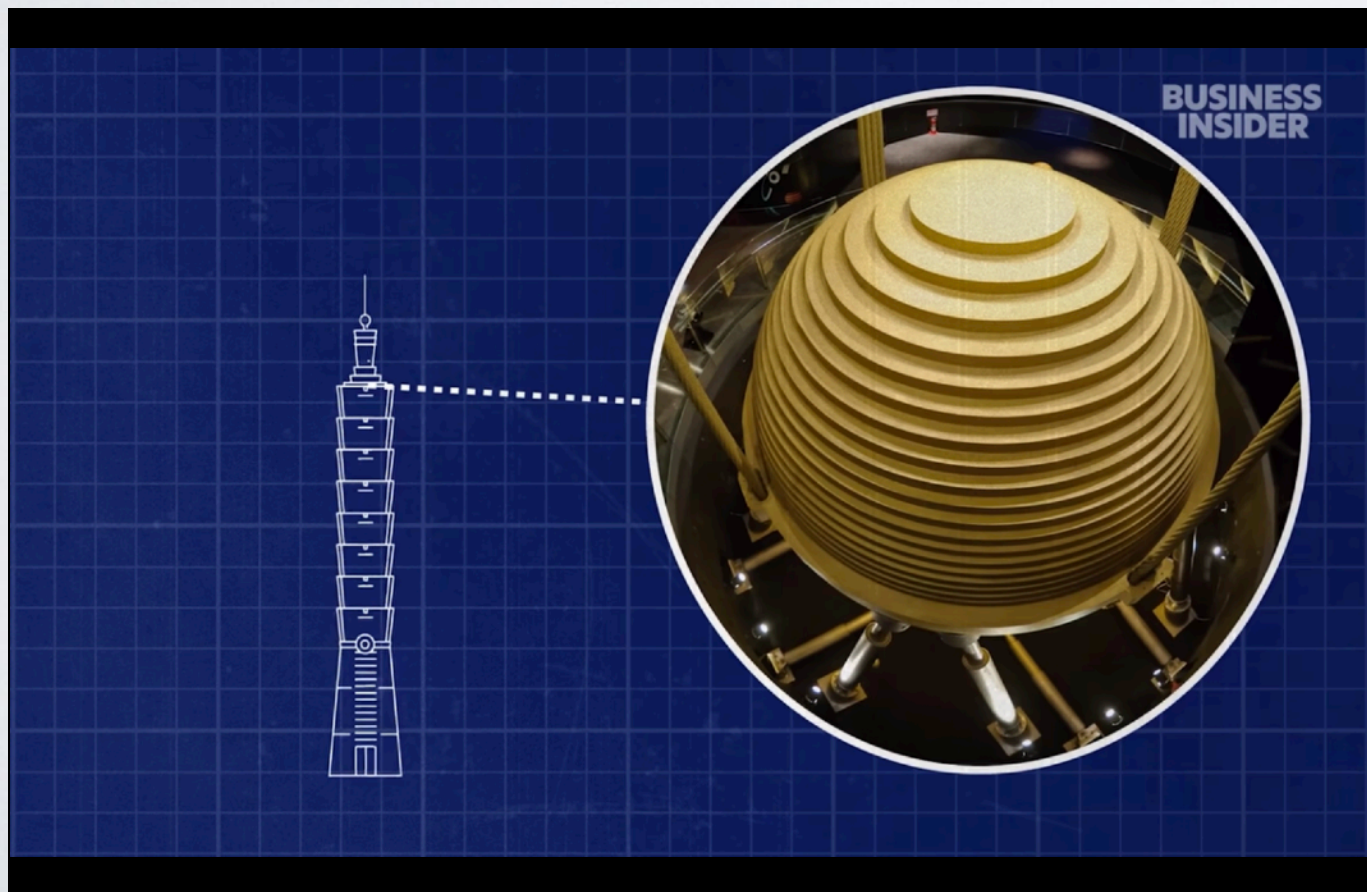
Improving the efficiency of algorithms for fundamental computations can have a widespread impact, as it can affect the overall speed of a large amount of computations. Matrix multiplication is one such primitive task, occurring in many systems—from neural networks to scientific computing routines. The automatic discovery of algorithms using machine learning offers the prospect of reaching beyond human

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Damper in Taipei 101, [youtube.com/watch?v=9DaEmvKWSEa](https://www.youtube.com/watch?v=9DaEmvKWSEa)

Grading guideline

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Damper in Shanghai Tower (1000t, using oil damper + eddy currents), Wiki commons

Grading guideline

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youtube.com/watch?v=9DaEmvKWSEa

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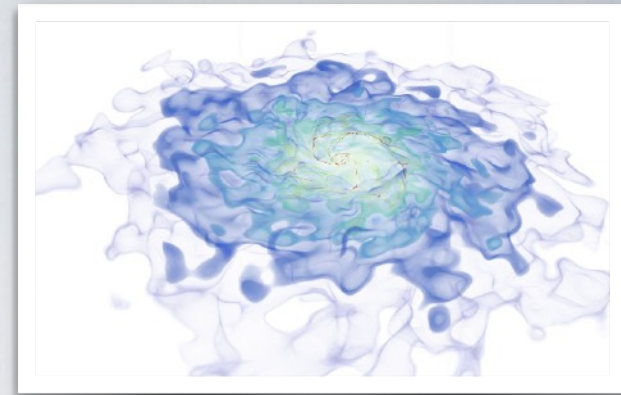
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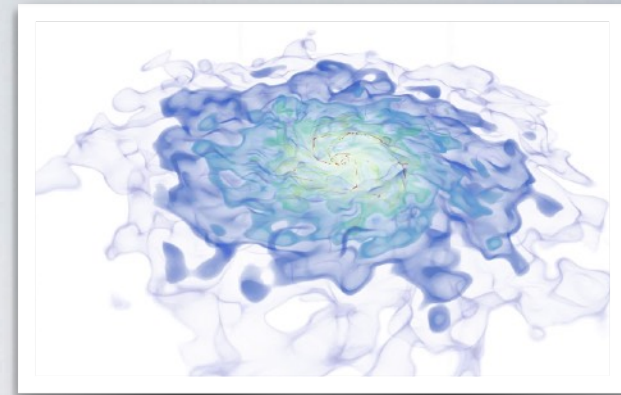


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In the first page, please write down
the total time you spent to prepare
your report!



Quantity Doesn't Guarantee Quality!

Student Honor Code

본 과목을 수강하는 학생들은 양심과 책임의식에 기반하며 시험, 과제물 및 보고서 작성 시에 학문의 정직성을 훼손하지 않겠다는 서약을 한 것으로 간주함

“본 과목을 수강하는 학생들은 양심과 책임의식에 기반하여 시험, 과제물 및 보고서 작성시에 학문의 정직성을 훼손하지 않겠다는 서약을 한 것으로 간주함.”

강의 수강 관련

- 시각장애: 교재 제작(디지털교재, 점자교재, 확대교재 등), 대필도우미 허용
- 지체장애: 교재 제작(디지털교재), 대필도우미 및 수업보조 도우미 허용
- 청각장애: 대필 및 문자통역 도우미 활동 허용, 강의 녹취 허용
- 건강장애: 질병 등으로 인한 결석에 대한 출석 인정, 대필도우미 허용
- 학습장애: 대필도우미 허용
- 지적장애/자폐성장애: 대필도우미 및 수업 멘토 허용

과제 및 평가 관련

- 시각장애/지체장애/청각장애/건강장애/학습장애: 과제 제출기한 연장, 과제 제출 및 응답 방식의 조정, 평가 시간 연장, 평가 문항 제시 및 응답 방식의 조정, 별도 고사실 제공
- 지적장애/자폐성장애: 개별화 과제 제출 및 대체 평가 실시

비고

본 강의를 수강하는 장애학생들에게는 이상의 지원 서비스 이외에도 장애학생 개개인의 특성과 요구에 따라, 지도교수 및 장애학생지원센터와의 상담을 통하여 적절한 수준의 지원 서비스를 제공합니다. 장애학생에 대한 지원서비스와 관련하여 문의사항이 있는 학생들은 담당교수 김지후(02-890-1210) 혹은 장애학생지원센터(02-890-8727)로 문의바랍니다.

AI Policy + SNU AI Guideline

*4. 생성형 AI 도구 활용 방침 ※ 필요에 따라 내용 수정 가능 교수자별 강의계획서 생성형 AI 도구 활용 방침	■ 본 강의 중 제출하는 모든 과제물(숙제, 퀴즈, 팀 프로젝트, 그 외 교수자가 학기중 지정하는 일체의 과제)은 생성형 AI를 활용하지 않고 작성/제출하며, 교수자와 조교는 이를 전제로 평가합니다. 이는 수강생의 독창적인 문제 해결력과 비판적 사고력 함양을 위한 정책입니다. ■ 다만, 과제 수행에 필요한 자료 검색 및 정리 등 제한적인 목적으로 생성형 AI를 사용할 수 있으며, 그럼에도 불구하고 제출된 과제물은 수강생 자신의 생각을 담은 자신의 언어로 작성되어야 합니다. 또한, 생성형 AI를 과제 수행에 어떤 용도라도 사용한 경우, 사용 도구명, 명령 프롬프트 및 생성 내용 등을 과제물에 명시해야 하며, 이렇게 제출된 과제물에 따르는 모든 책임 (학문의 진실성을 증명할 의무 및 저작권 침해를 방지할 의무 포함)은 본인에게 있음을 인지해야 합니다. 자신의 생각을 반영하여 수정/보완이 되지 않은 AI 산출물을 제출하는 것은 학문의 진실성을 위반한 것으로 평가될 수 있습니다. ■ 그 외에 여기에 명시되지 않은 사항은 "서울대학교 AI 가이드라인"(2025.01)을 준용합니다.
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■ All assignments in this course (including homework, quizzes, term projects, and any other assignments designated by the instructor) must be prepared and submitted without the use of generative AI, and the instructor and TAs will evaluate them under this assumption. This policy is intended to ensure that students develop and demonstrate their own creative problem-solving skills and critical thinking abilities, without reliance on generative AI tools.

■ Generative AI may, however, be used for limited purposes, such as searching for and organizing materials necessary for assignments. Even in such cases, all submitted work must be entirely the student's own and written in the student's own words, reflecting the student's independent thought process. If generative AI is used for any purpose during the completion of an assignment, students must clearly disclose the name of the tool used, the prompts provided, and the generated content. Students must also acknowledge that they bear full responsibility for any consequences arising from such use, including the obligation to uphold academic integrity and to prevent copyright infringement. Submitting AI-generated output without substantial revision or meaningful incorporation of the student's own ideas may be considered a violation of academic integrity.

■ For matters not explicitly addressed here, the "SNU AI Guidelines" (January 2025) shall apply.

- Your term paper must include:

1. Your own **originality**

2. (Optional) Appropriate use of computing or AI, for either
(a) a **validated** speed-up, or, (b) data-driven **discovery**

1 Do something original

Don't just summarize or repeat what others have done.

Your paper should include **your own idea, analysis, or way of thinking about a problem.**

2 (Optional) Use computing or AI — but use it properly

You can use coding or AI tools, but only if they are used in a **meaningful and responsible way.**

There are two good ways to use them:

- **(a) Validated speed-up**

You use computing or AI to do something faster (e.g., calculations, simulations),
but you must check and understand the results yourself.

→ Don't just trust the output—make sure it's correct.

- **(b) Data-driven discovery**

You use data, simulations, or AI to **find patterns, insights, or new results,**
and explain clearly how you got them.

⊘ What this is really saying

- Don't just copy or summarize → **add your own thinking**
- Don't just use AI blindly → **understand and verify what it does**

ChatGPT's
understanding of the
term paper guideline

- Your term paper must include:

1. Your own originality

2. (Optional) Appropriate use of computing or AI, for either
(a) a validated speed-up, or, (b) data-driven discovery

서울대학교 과학데이터혁신연구소
특별세미나
AI for Science:
복잡한 과학데이터를 어떻게 이해할 것인가?
Masterclass in Token Cartography: Decoding Complex Scientific Data with AI for Science

2026. 9. 18.(금) 오후 2시-4시
서울대학교 25동 405호

연사
Prof. N. Duane LOH
National University of Singapore
Department of Physics &
Department of Biological Sciences
Program Director, MSc in AI for Science

QR code: 2026. 9. 18. (금) 오후 2시-4시

Scientific instrumentation is generating noisy, high-dimensional, and complex data at an unprecedented rate, surpassing the capacity for interpretation. The primary challenge, I claim, lies in finding the few key dimensions in which these data vary meaningfully; the remaining dimensions exhibit relatively unimportant variations and uncertainties. The observation that many collected datasets fall into a limited number of such dimensions is known as the manifold hypothesis. Unsupervised and semi-supervised dimension reduction, widely established, effectively isolates these few directions and presents the data in a comprehensible manner for scientists. Effectively, this reduction yields a latent space of variations that explains the diversity in most of the data. Despite this progress, navigating the low-dimensional space remains challenging. Participants tokenize this latent space using vector quantization, a method with independent roots in statistics, mathematics, information theory, and signal processing. The idea is to divide this space into patches, each representing the measurements contained within. This deliberate decision involves discarding variation within a token, allowing scientists to trace changes that hold explanatory significance. This action is inextricably linked to the concept of coarse-graining in statistical physics — essentially deciding which degrees of freedom can be traced out, and which phenomenological parameters emerge. These tokens also allow us to navigate the complex latent spaces occupied by our observations. Over a span of two hours, participants engage with real datasets through interactive dashboards, without coding. They experiment with different compression and partitioning strategies, identify discrepancies between the map and their existing knowledge, and formulate hypotheses based on these disagreements. Subsequently, they propose experiments to resolve these discrepancies, recognizing that devising a robust test is the more challenging and informative skill.

주최 서울대학교 과학데이터혁신연구소 후원 SNU G-LAMP
Institute for Data Innovation in Science Seoul National University 서울대학교 G-영역사업단

- Your term paper must include:

1. Your own originality

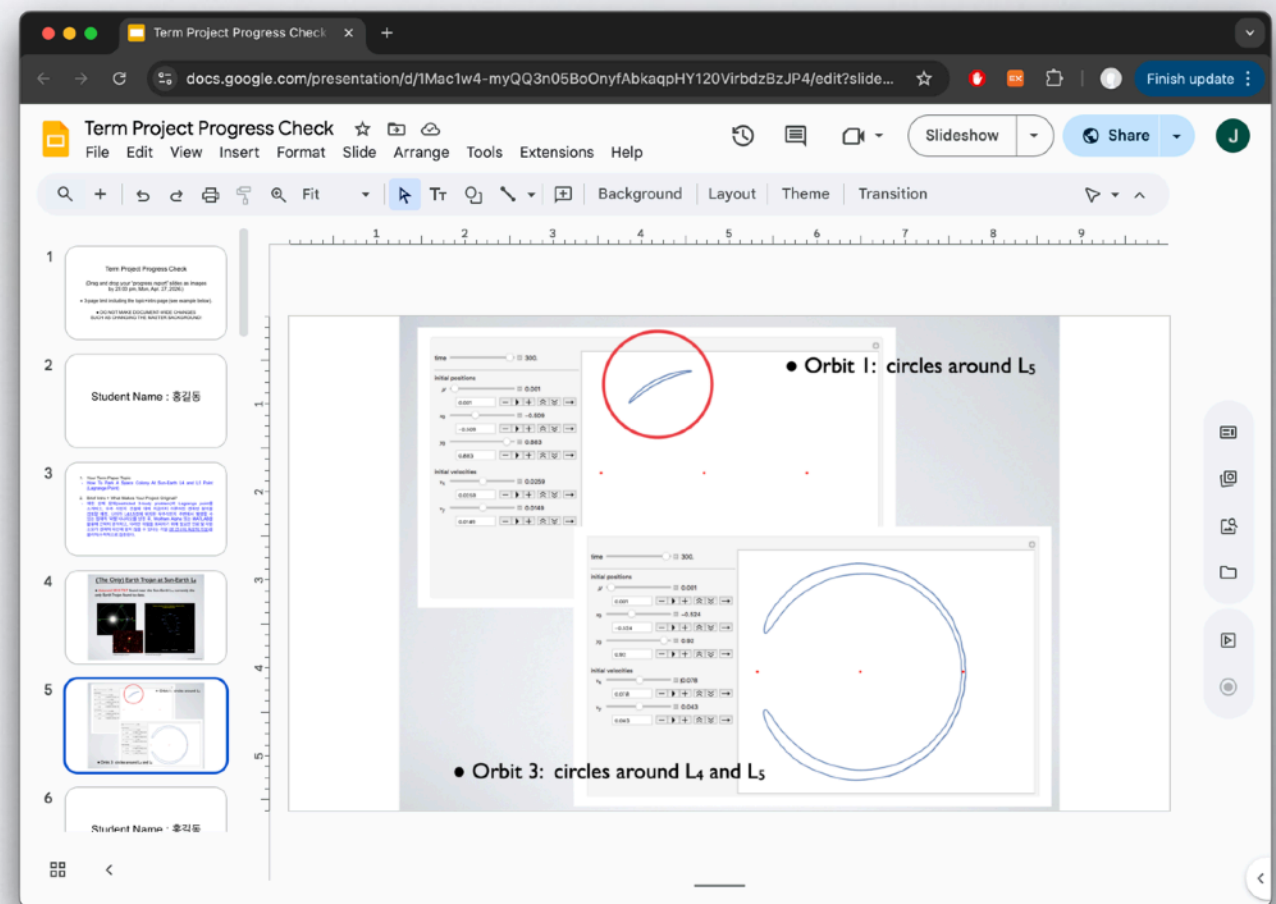
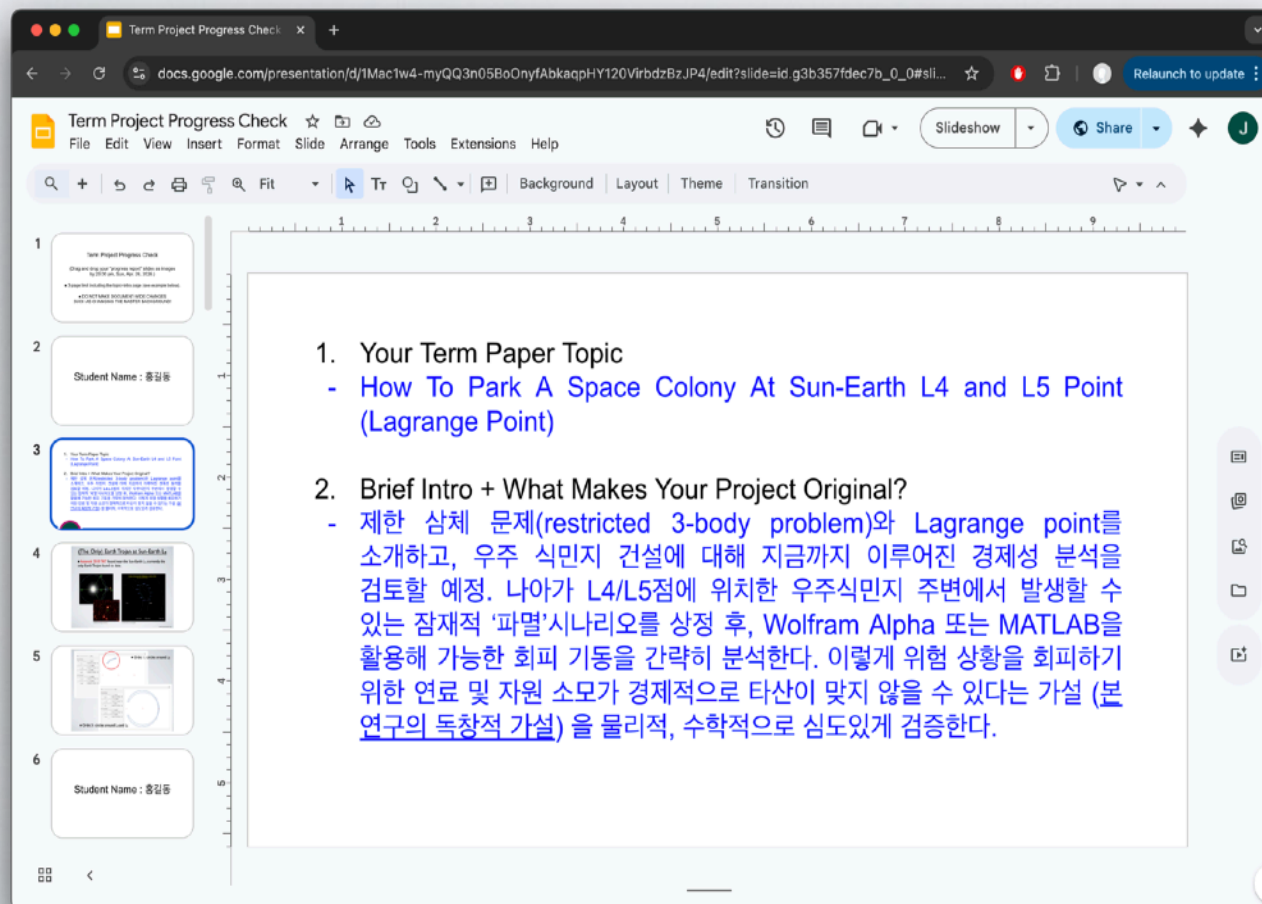
2. (Optional) Appropriate use of computing or AI, for either
(a) a validated speed-up, or, (b) data-driven discovery



Term Project: Topic & Progress Check!

- **Post your topic** (+ answer to “What makes your project original?”) and **2-page “progress report” slides** as images **by 23:00 pm, Oct. 27 (Tue)** on a shared Google doc:

https://docs.google.com/presentation/d/1t2kbpGYGkcmJx5QXF4JSP7JYIxjvIkawlr bmpU_2k0



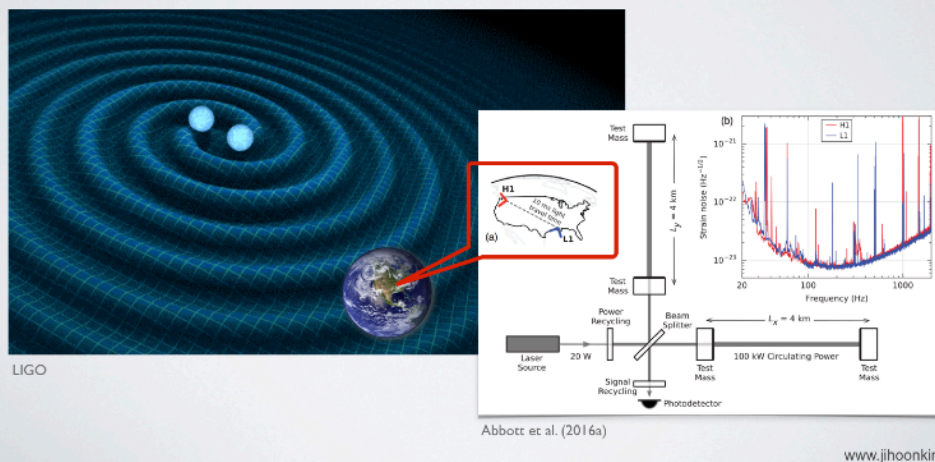
Term Project: Topic & Progress Check!

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https://docs.google.com/presentation/d/1t2kdbpGYGkcmJx5QXF4JSP7JYIxjvIkawlr bmpU_2k0

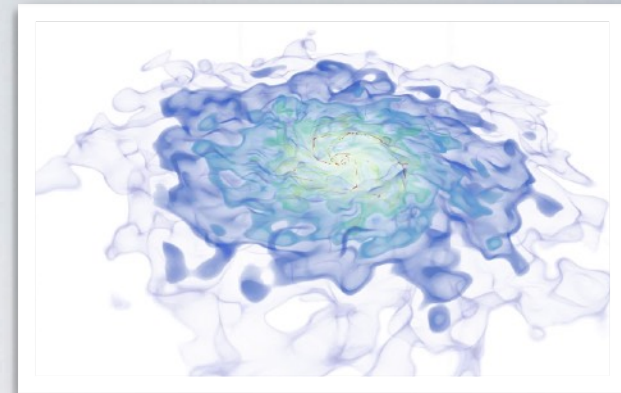
LIGO Gravitational Wave Signals

- Signals from distant events that distorted the spacetime fabric.
- **A new window to the Universe** opened up in addition to the good old electromagnetic window.



Extra credit guideline

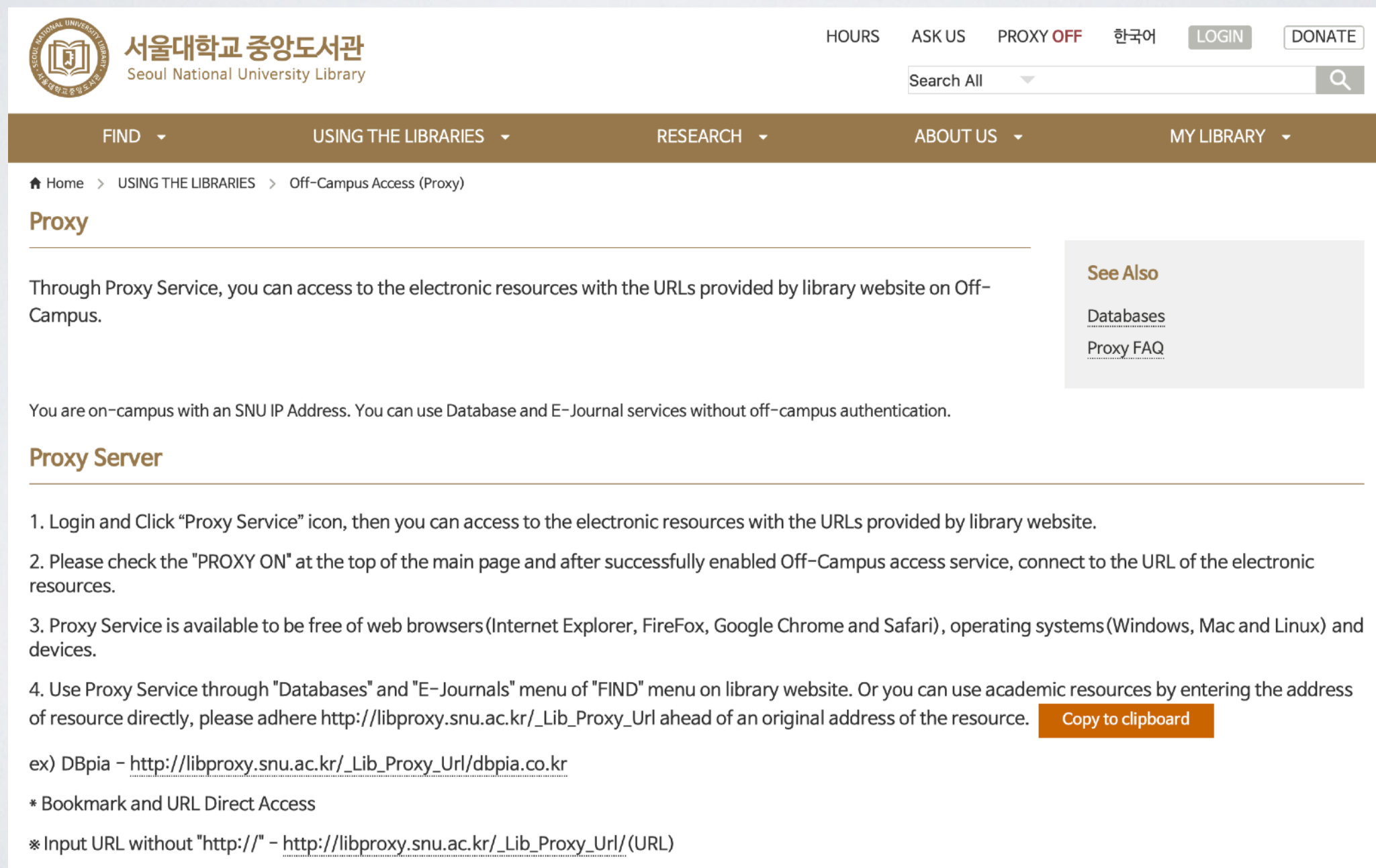
- **Additional +2:** awarded to 2-3 students who were voted the best by your peers, and to 2-3 students who provided the most useful feedback.



Supplemental Slides

Electronic Resources via SNU Library

- To access e.g., academic journals off-campus via SNU library's proxy service, visit library.snu.ac.kr/using/proxy.



The screenshot shows the Seoul National University Library website. The header includes the library logo, name in Korean and English, and navigation links for HOURS, ASK US, PROXY OFF, 한국어, LOGIN, and DONATE. A search bar is also present. The main navigation menu has categories: FIND, USING THE LIBRARIES, RESEARCH, ABOUT US, and MY LIBRARY. The breadcrumb trail indicates the current location: Home > USING THE LIBRARIES > Off-Campus Access (Proxy). The page title is "Proxy". The main content explains that the Proxy Service allows access to electronic resources with library-provided URLs on an off-campus. It notes that users on-campus with an SNU IP address can use database and e-journal services without authentication. A "See Also" sidebar lists "Databases" and "Proxy FAQ". The "Proxy Server" section provides a four-step guide: 1. Login and click "Proxy Service". 2. Check "PROXY ON" at the top of the main page. 3. Note that the service is available on various web browsers and operating systems. 4. Use the service through "Databases" and "E-Journals" or directly by adding the proxy URL to the resource address. An example is given for DBpia. A "Copy to clipboard" button is provided for the example URL. Additional instructions include bookmarking the direct access URL and a note about inputting the URL without "http://".

서울대학교 중앙도서관
Seoul National University Library

HOURS ASK US PROXY OFF 한국어 LOGIN DONATE

Search All

FIND USING THE LIBRARIES RESEARCH ABOUT US MY LIBRARY

Home > USING THE LIBRARIES > Off-Campus Access (Proxy)

Proxy

Through Proxy Service, you can access to the electronic resources with the URLs provided by library website on Off-Campus.

You are on-campus with an SNU IP Address. You can use Database and E-Journal services without off-campus authentication.

Proxy Server

1. Login and Click "Proxy Service" icon, then you can access to the electronic resources with the URLs provided by library website.
2. Please check the "PROXY ON" at the top of the main page and after successfully enabled Off-Campus access service, connect to the URL of the electronic resources.
3. Proxy Service is available to be free of web browsers (Internet Explorer, FireFox, Google Chrome and Safari), operating systems (Windows, Mac and Linux) and devices.
4. Use Proxy Service through "Databases" and "E-Journals" menu of "FIND" menu on library website. Or you can use academic resources by entering the address of resource directly, please adhere http://libproxy.snu.ac.kr/_Lib_Proxy_Url ahead of an original address of the resource. [Copy to clipboard](#)

ex) DBpia - http://libproxy.snu.ac.kr/_Lib_Proxy_Url/dbpia.co.kr

* Bookmark and URL Direct Access

※ Input URL without "http://" - [http://libproxy.snu.ac.kr/_Lib_Proxy_Url/\(URL\)](http://libproxy.snu.ac.kr/_Lib_Proxy_Url/(URL))

See Also

- Databases
- Proxy FAQ

Resources via SNU Liberal Education

- To access e.g., academic journals off-campus via SNU library's proxy service, visit library.snu.ac.kr/using/proxy.

The image displays two overlapping browser windows showing SNU Liberal Education resources.

Left Window (liberaledu.snu.ac.kr/글쓰기-클리닉/):

- Header: 서울대학교 기초교육원 (Faculty of Liberal Education)
- Navigation: 교양교육과정, 자율연구, 교육·학생지원, 교수학습센터, 글쓰기센터, 정보
- Section: 글쓰기 프로그램 (Writing Program)
- Sub-section: 글쓰기 클리닉 (Writing Clinic)
- Content: 글쓰기 능력 향상 프로그램 (Writing Ability Improvement Program)
- Text: 방학 기간 중 튜터와의 일대일 대면 지도를 통해 자신의 글쓰기 능력을 향상시킬 수 있도록 마련된 프로그램입니다. 튜터로부터 자신의 글쓰기 습관에 맞는 글쓰기 과제를 부여 받아 작성하는 과정을 통해 글쓰기에 대한 자신감을 키울 수 있습니다.
- Text: ※ 이 프로그램을 성실하게 이수한 학생에게는 이수증이 발급됩니다.
- Text: · 신청 대상: 학부 및 대학원생(외국인 학생은 '외국인 글쓰기 지도 프로그램'을 이용해 주세요.)
- Text: · 신청 방법: 서울대학교 포털 홈페이지(<https://my.snu.ac.kr>) → 상단의 '비교과'(<https://extra.snu.ac.kr>) → '글쓰기 능력향상 프로그램' 검색

Right Window (owl.snu.ac.kr/글쓰기-길잡이/):

- Header: 온라인 글쓰기교실 (ONLINE WRITING LAB)
- Navigation: 글쓰기 길잡이, 아카이브, 블로그, 공지사항, 추천 링크, 글쓰기 FAQ
- Section: 글쓰기 길잡이 (Writing Guide)
- Text: 서울대학교 온라인 글쓰기교실 글쓰기 길잡이 코너입니다
- Text: 글쓰기가 막막할 때 글쓰기의 시작부터 마무리까지 도움이 되는 길잡이로 참고하실 수 있습니다
- Grid of Topics:

글쓰기 시작하기	글쓰기 전개하기	글쓰기 마무리하기	종류별 글쓰기
→ 대학에서의 글쓰기	→ 논증하기	→ 인용하기와 인용 출처 표기하기	→ 논문 쓰기
→ 과제 이해하기	→ 설명하기	→ 수정하기	→ 서평 쓰기
→ 주제 찾기	→ 문제해결하기	→ 마무리하기	→ 학술 에세이 쓰기
→ 주제문과 개요 작성하기			→ 비평적 글쓰기

Footer: <https://liberaledu.snu.ac.kr>